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CLIENT HANDOVER DOCUMENT

ENTERPRISE SOLUTIONS → PRODUCTION PLATFORM

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3D GALAXY E-COMMERCE & CUSTOM MANUFACTURING PLATFORM

ADMIN PORTAL

STANDARD OPERATING PROCEDURE (SOP)

Complete Administration, Configuration & Operations Guide for Physical Retail Catalog, Dynamic Multi-Group Variants, Custom Additive Manufacturing (STL) Slicing Quotes, Omnichannel Communications & Disaster Resilience.

DELIVERED BY

AJR Digital Hub Team

Enterprise Digital Solutions Team

DELIVERED TO / ACCEPTED BY

3D Galaxy Team

E-Commerce Operations & Store Administration

DOCUMENT RELEASE

Version 1.0 (Production Master)

Published: September 2026

CLASSIFICATION

Commercial in Confidence

Client Operations Manual

DOCUMENT CONTROL & GOVERNANCE



DELIVERING PARTNER
AJR Digital Hub Team

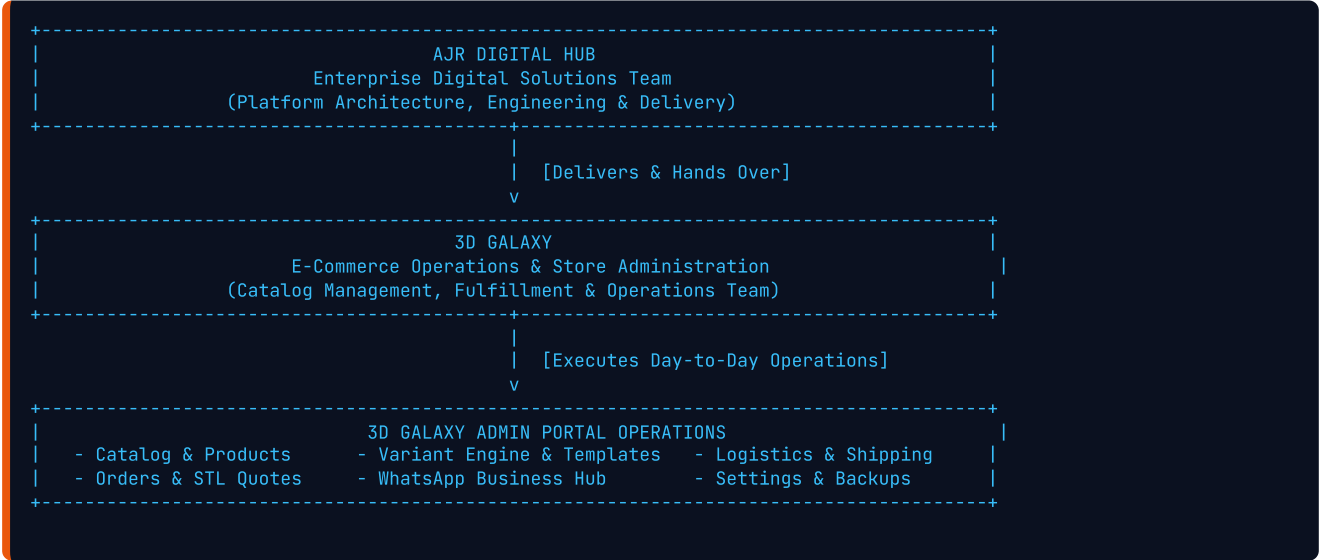
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RECEIVING CLIENT
3D Galaxy Team



DOCUMENT ATTRIBUTE	SPECIFICATION DETAILS
Document Title	3D Galaxy Admin Portal — Standard Operating Procedure (SOP)
Application Platform	3D Galaxy Enterprise E-Commerce & Custom Manufacturing Portal
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Release Version	1.0 (Production Master)
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Classification	Commercial in Confidence — Client Operational Manual
Delivered By	AJR Digital Hub Team (Enterprise Digital Solutions Team)
Delivered To / Accepted By	3D Galaxy Team (E-Commerce Operations & Store Administration)
Target Audience	Store Administrators, Inventory Managers, Fulfillment Officers, Support Leads

Document Handover Relationship



Document Revision History

VERSION	RELEASE DATE	AUTHOR / ORGANIZATION	SUMMARY OF REVISIONS
0.1	Sep 18, 2026	AJR Digital Hub Engineering	Initial portal audit, navigation route inventory, data model extraction

0.5	Sep 18, 2026	AJR Digital Hub Solutions	Workflow validation, variant matrices, WhatsApp Cloud API architecture
1.0	Sep 19, 2026	AJR Digital Hub Delivery Team	Complete client handover manual with AJR branding, 30 figures, 11 product tabs, 24 settings sub-tabs, and 20 use cases

EXECUTIVE SUMMARY

The **3D Galaxy Admin Portal** is an enterprise management dashboard engineered and delivered by **AJR Digital Hub** for the **3D Galaxy** team to orchestrate dual-track e-commerce operations: high-velocity physical inventory sales and on-demand custom additive manufacturing (3D printing) services. By coupling standard retail catalog workflows with specialized slicing parameter analysis, the platform provides complete lifecycle control from customer cart placement to automated courier tracking dispatch.

Core Business Pillars

SYSTEM MODULE	PRIMARY OPERATIONAL FUNCTION	DIRECT BUSINESS VALUE
Catalog & Variant Engine	Multi-attribute combinations, color swatches, dynamic bundle builder, instant templates.	Enables complex SKU configurations (Material × Weight × Color) in under 60 seconds without database overhead.
Logistics & STL Slicing	Real-time order inspection, offline sales dockets, custom 3D model geometry analysis (volume, mass, layer height).	Eliminates manual quoting bottlenecks; allows automated volume-to-price conversion for on-demand fabrication.
Conversational Commerce	Meta WhatsApp Cloud API integration with rule-based auto-replies, customer emoji reactions, and admin attachments.	Achieves 98% message open rate; provides instant customer service with zero recurring AI token consumption fees.
Push Engagement Hub	Broadcast notification builder, deep-link routing, rich imagery, and automated abandoned cart recovery triggers.	Recuperates up to 22% of abandoned checkouts through automated discount voucher blasts.
Disaster Recovery Engine	Weekly scheduled database backups, cryptographic SHA-256 integrity hashing, point-in-time restoration.	Ensures zero data loss and business continuity in the event of hardware or infrastructure failure.

How to Use This SOP (Role-Based Guide)

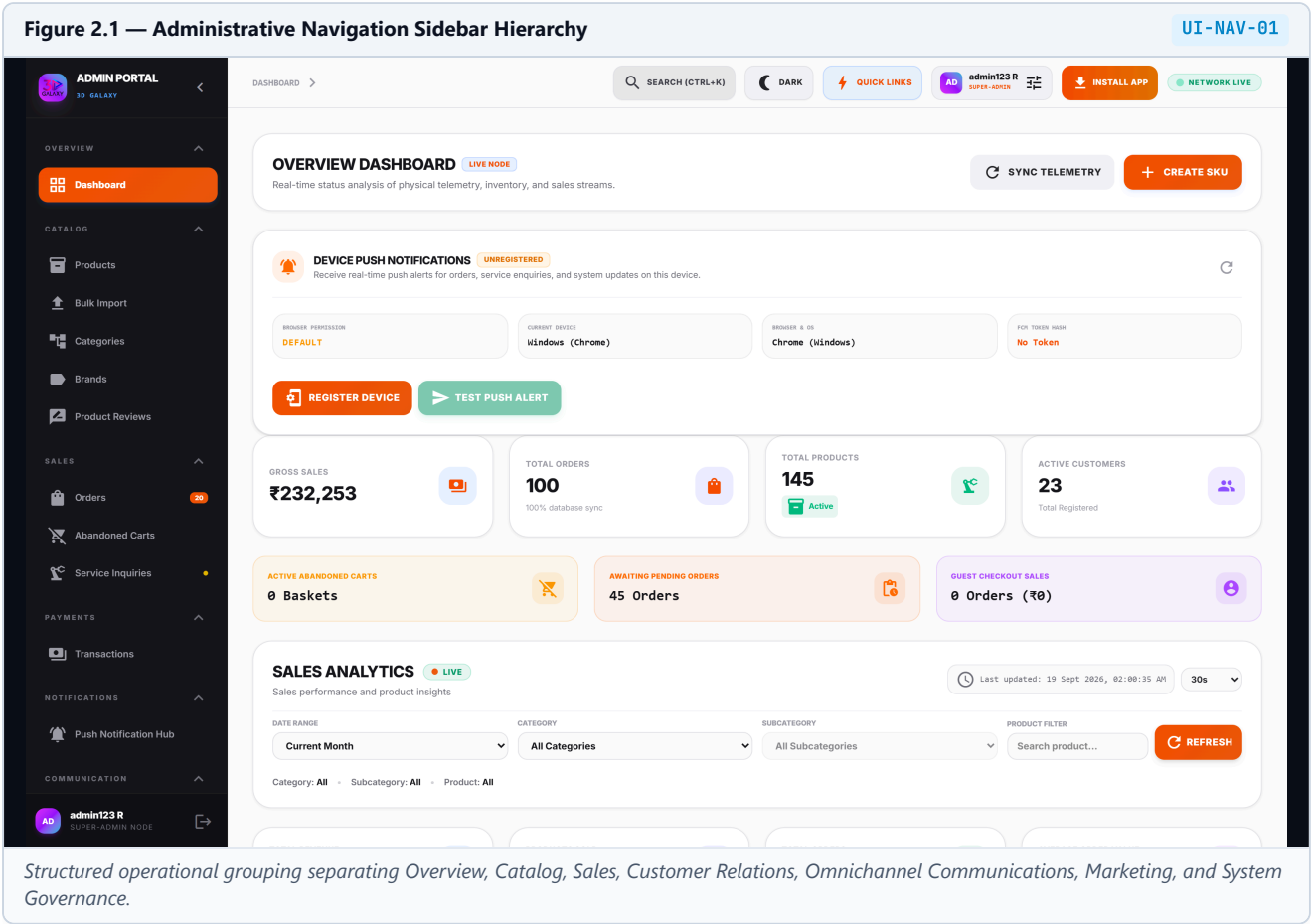
OPERATIONAL ROLE	RECOMMENDED STARTING CHAPTERS & DIRECT ACTIONS
New Administrator	Start with Section 1 (Access Security) , Section 2 (Navigation Hierarchy) , and Section 3 (Dashboard Overview) .
Catalog & Product Manager	Review Section 4 (Catalog Overview) , master Section 5 (11 Product Tabs) , and implement Section 6 (Variant Guide & Templates) .
Fulfillment & Logistics Lead	Execute Section 8 (Orders Console) , Section 9 (STL Slicing Quotes) , Section 10 (Packing Slips) , and Section 19.17 (Shipping Rules) .
Customer Support & Marketing	Operate Section 14 (WhatsApp Inbox) , Section 15 (WhatsApp Rules) , Section 16 (Push Notifications) , and Section 11 (CRM Directory) .
Platform Administrator	Configure global settings in Section 19 (All 24 Settings Sub-Tabs) and manage security tokens.
Disaster Recovery Lead	Follow procedures in Section 20 (Database Backup, Cryptographic Hashing & Restoration) .

OPERATIONAL SECURITY & COMPLIANCE MANDATE

Administrative personnel must observe strict security protocols. Never disclose administrative credentials, API secrets, or customer PII over unencrypted channels. Ensure role-based access delegation is enforced across all terminal sessions.

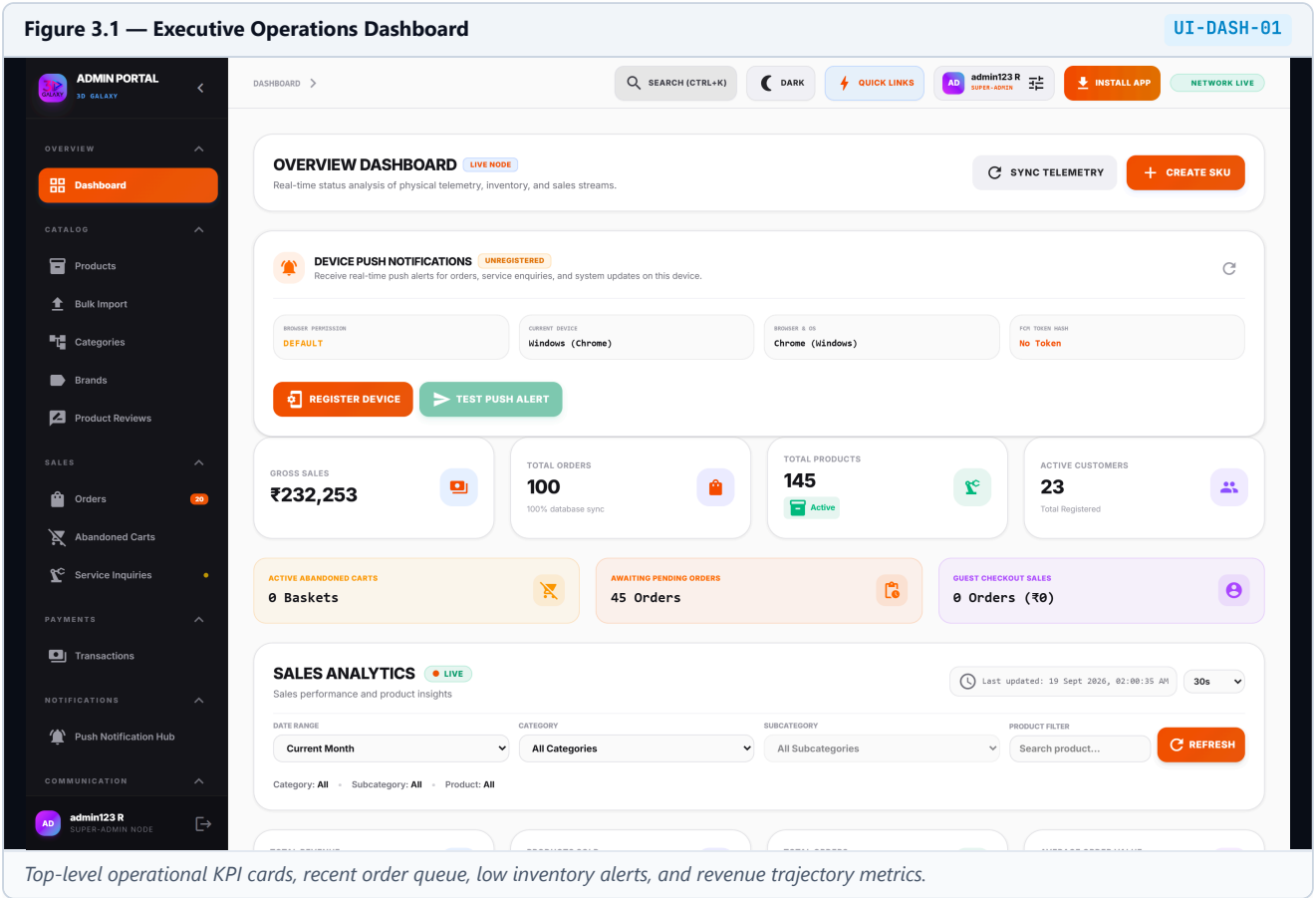
ADMIN PORTAL NAVIGATION HIERARCHY

The navigation bar is organized logically on the left sidebar (and responsive mobile drawer) into 7 functional operational groups, ensuring rapid switching between sales, catalog, and communication tasks.



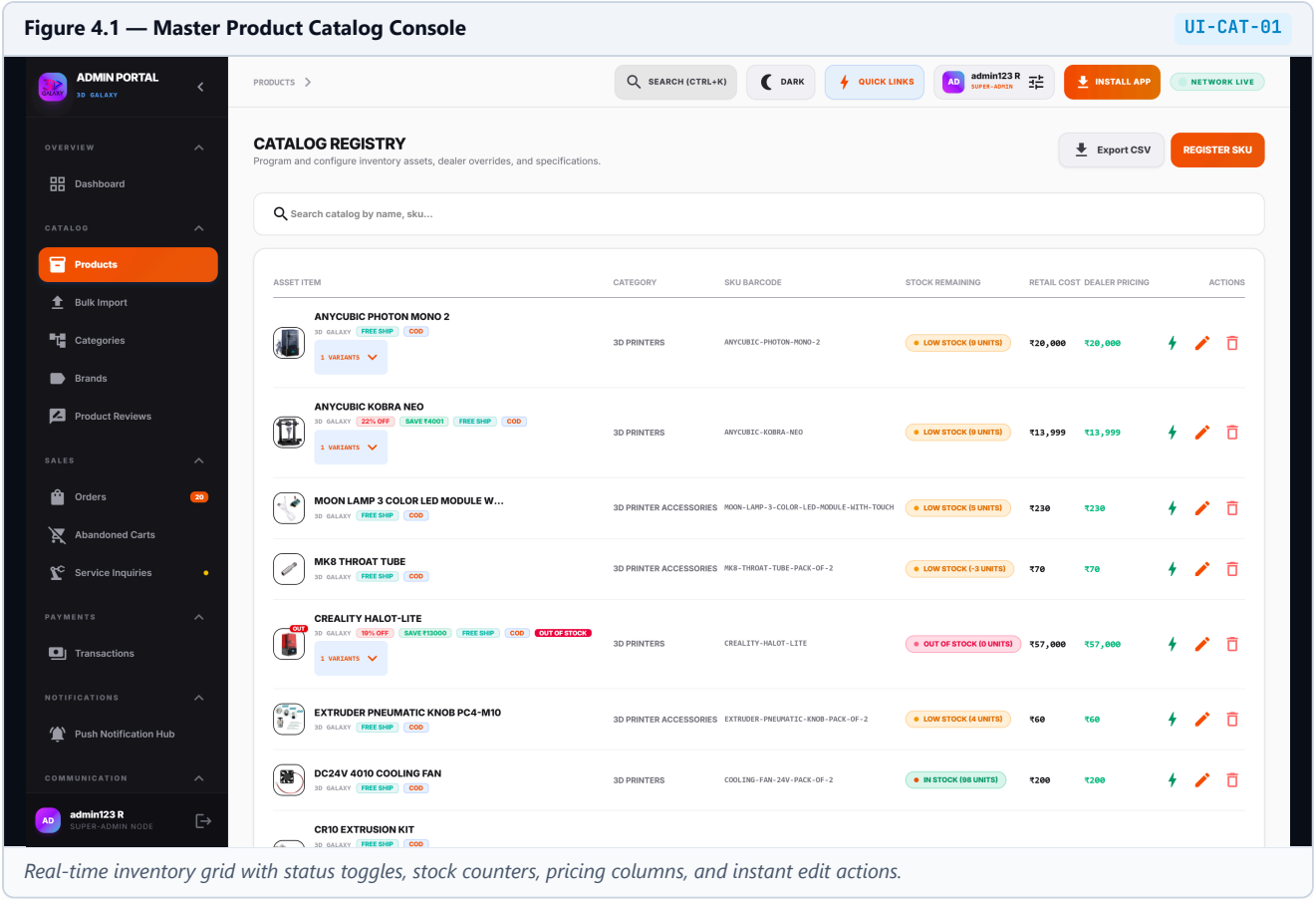
EXECUTIVE DASHBOARD & OPERATIONS HUB

The Executive Dashboard serves as the operational cockpit for store supervisors, providing real-time telemetry on revenue velocity, fulfillment pipeline health, inventory replenishment alerts, and customer engagement.



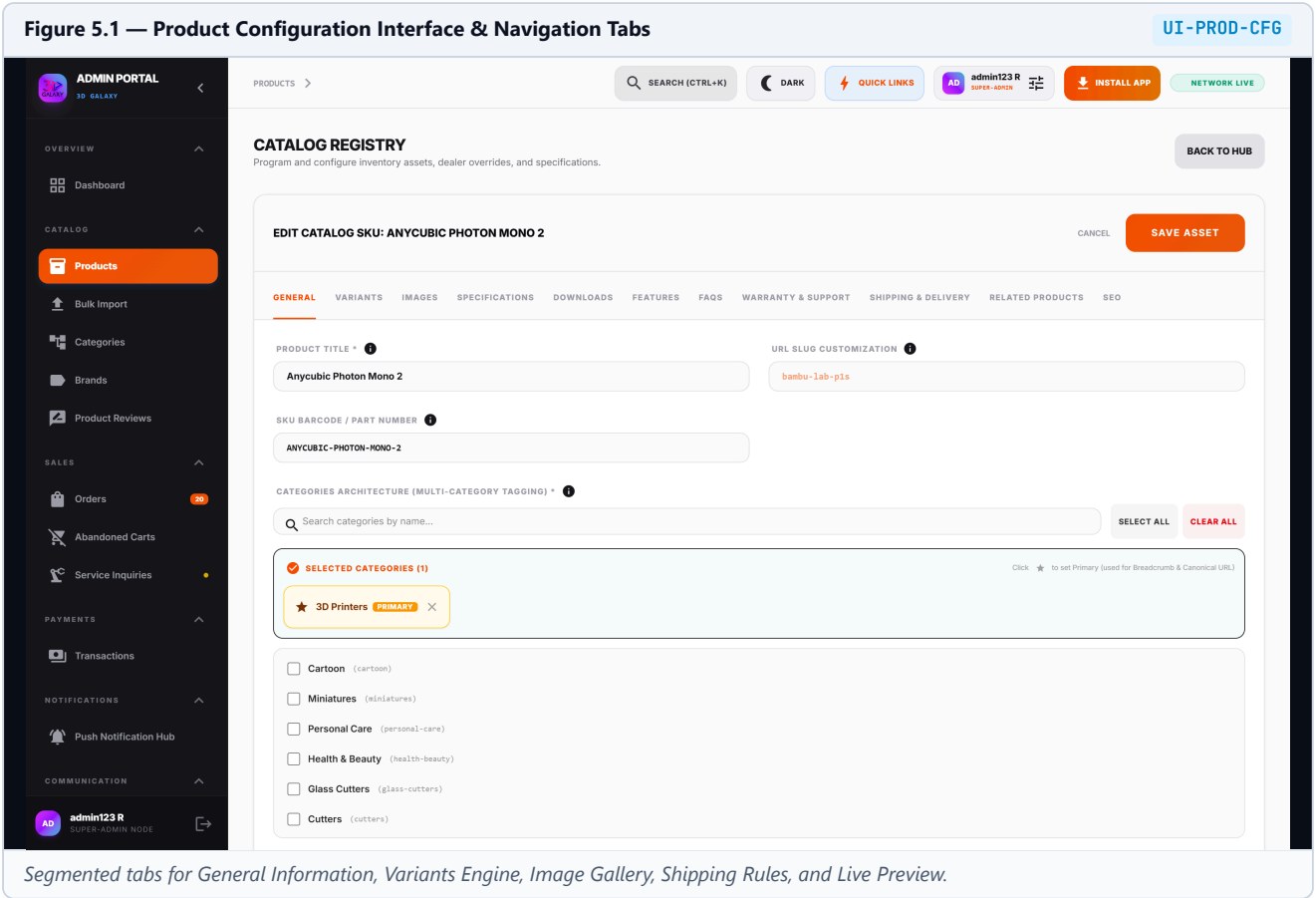
Master Product Catalog Console

The Catalog view renders physical inventory items, SKU identifiers, categories, retail pricing, wholesale dealer prices, and live publication status.



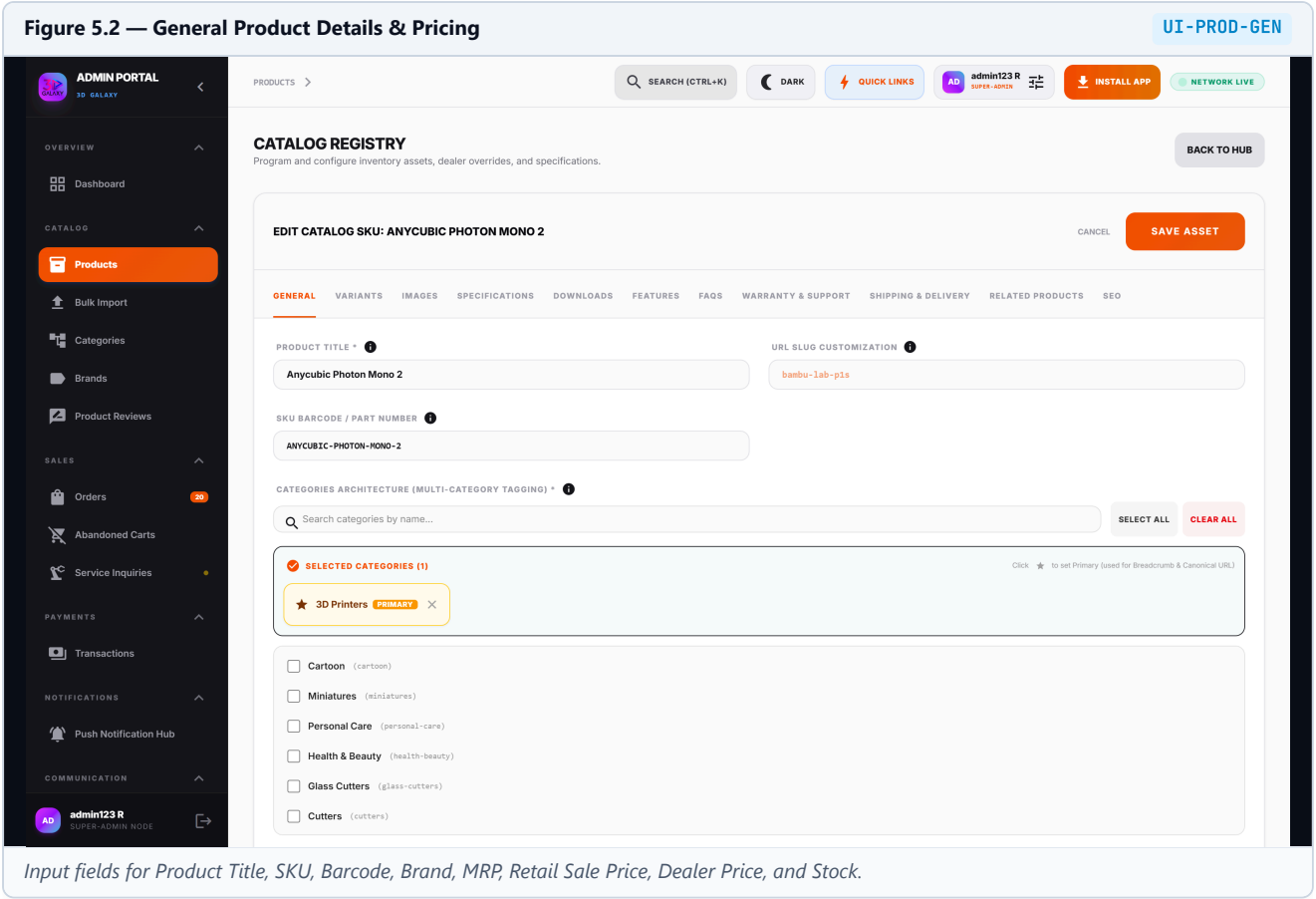
PRODUCT CONFIGURATION: THE 11 FUNCTIONAL TABS

The Product Editor consists of **11 specialized tabs**. Each tab governs a distinct functional aspect of customer presentation, warehouse logistics, and SEO discoverability.



1. General Details

Defines the canonical catalog identity, SKU code, category assignment, base pricing, warehouse stock, and status policy.



FIELD	OPERATIONAL IMPACT OF VALUE CHANGES
Sale Price (₹)	Modifies checkout line total, cart tax calculation, and payment gateway charge amount.
Physical Stock	Controls in-stock badge. When 0, disables Add-to-Cart unless backorders are enabled.
Category	Determines storefront menu navigation, filter faceted search, and category-level shipping fees.
Status Policy	Active Storefront (public), Draft (admin only), or Out of Stock (visible but unpurchasable).

The Remaining 10 Configuration Tabs

TAB NAME	AVAILABLE FIELDS & ADMIN CONTROLS	CUSTOMER & BUSINESS IMPACT
2. Variants	Group builder, display types (11 modes), selection modes (6 rules), value chips, bundle tiers.	Enables color, size, weight, and multi-pack options directly on product page.
3. Images	Hero photo uploader, multi-angle gallery, and Linked Variant Dropdown per photo.	Automatically changes main gallery photo when customer clicks a specific color chip.
4. Specifications	Key-value technical attributes (Nozzle size, build volume, bed temp, layer height).	Renders technical comparison table on storefront product specifications tab.
5. Downloads	Slicing profiles (`.curaprofile`, `.3mf`), firmware binaries (`.bin`), user manuals (`.pdf`).	Provides instant post-purchase self-service downloads, reducing support tickets.
6. Features	Visual feature cards with icons, bold titles, and technical highlight blurbs.	Highlights core selling propositions directly beneath the buy box.

7. FAQs	Product-specific accordion Q&A pairs with sort order.	Resolves pre-purchase compatibility and maintenance questions, boosting conversion.
8. Warranty & Support	Warranty duration (12/24 mos), support channels (Email/WhatsApp), DOA replacement terms.	Provides transparent buyer assurance and legal compliance under consumer guidelines.
9. Shipping & Delivery	Custom shipping fee, free shipping toggle, calibrated weight in grams, delivery days buffer.	Drives courier tariff calculations and dynamic "Expected Delivery by [Date]" countdown.
10. Related Products	Multi-select cross-sell picker from live catalog.	Renders "Frequently Bought Together" carousel, driving Average Order Value (AOV).
11. SEO	Meta title, meta description, keywords, canonical URL, and OpenGraph social share card.	Ensures high search engine ranking and rich visual previews when links are shared on WhatsApp.

PRODUCT VARIANT CONFIGURATION GUIDE

The 3D Galaxy Variant Engine supports advanced multi-group architectures, allowing complex retail configurations such as **Material × Weight × Color** to be generated in seconds.

Figure 6.1 — Product Variant Architecture Builder

UI-VAR-BUILDER

ADMIN PORTAL

3D GALAXY

OVERVIEW

Dashboard

CATALOG

Products

Bulk Import

Categories

Brands

Product Reviews

SALES

Orders

Abandoned Carts

Service Inquiries

PAYMENTS

Transactions

NOTIFICATIONS

Push Notification Hub

COMMUNICATION

admin123 R

SUPER-ADMIN NODE

PRODUCTS

CATALOG REGISTRY

Program and configure inventory assets, dealer overrides, and specifications.

EDIT CATALOG SKU: ANYCUBIC PHOTON MONO 2

CANCEL

SAVE ASSET

GENERAL

VARIANTS

IMAGES

SPECIFICATIONS

DOWNLOADS

FEATURES

FAQS

WARRANTY & SUPPORT

SHIPPING & DELIVERY

RELATED PRODUCTS

SEO

VARIANT SETUP GUIDE & ONBOARDING

Step-by-step interactive walkthrough for options, pricing, stock, images, and live previews.

TAKE VARIANT CONFIGURATION TOUR

Help

VARIANT TEMPLATES

QUICK SETUP

Quickly configure common product variants (Filament Colors, Spool Weights, Bundles, Printers).

Choose Template

Product Variant & Bundle Group Architecture

Configure display types, selection modes, bundle pricing, duplicate rules, and test live rendering.

Add Variant Group

Title

General Settings

VARIANT NAME (INTERNAL)

DISPLAY NAME (FRONTEND LABEL)

Title

VARIANT TYPE & DISPLAY MODE

SELECTION MODE

Chip Selector (Pills)

Single Selection (Choose 1)

Required Field

Active

Allow Duplicate Selection in Slots

Admin interface for configuring variant groups, display types, selection rules, and duplicate slot allowances.

The 11 Verified Display Types

DISPLAY TYPE	RECOMMENDED USE CASE	CUSTOMER EXPERIENCE & ADMIN CONFIG
bundle-builder	Multi-pack bundles (Buy 1, Buy 3, Buy 5).	Interactive radio cards with custom color/item slot selectors below. Configure tiers and duplicate slot permissions.
weight-selector	Consumables (250g, 500g, 1kg, 2kg, 5kg).	Weight pill buttons calibrated directly with shipping calculations. Enter weight values in grams or kilograms.
chip	Text options (Sizes S, M, L, XL; Nozzle sizes).	Compact rounded pill chips. Admin enters comma-separated values; drag handles reorder chips.
dropdown	Extensive lists (> 10 items, spare parts).	Standard select dropdown menu saving screen space. Ideal for technical hardware selections.
image	Visual textures, machine finishes, colorways.	Grid of square visual thumbnails with active checkmark badge. Admin uploads image per option.
card	Machine models (Base, Combo AMS, Complete).	Rich cards with titles, bullet points, and pricing offsets. Ideal for high-ticket configurations.
radio-chips	Warranty plans, service add-ons.	Chip buttons with embedded circular radio indicators.

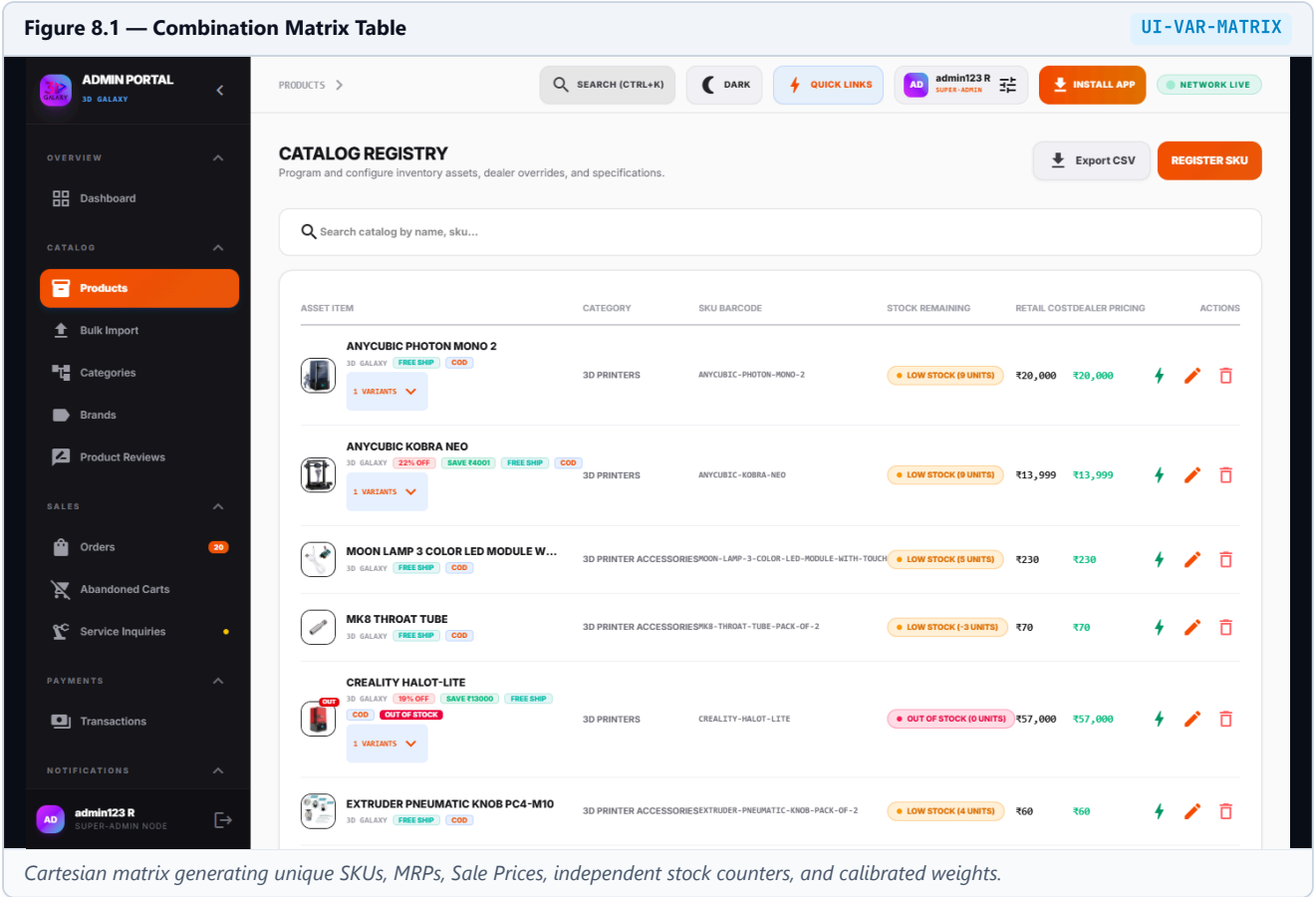
color-chips	Filament colors, resin hues.	Circular swatches showing exact hex codes (e.g. #FF2200) with tooltip name and active border.
button-group	Segmented choices (110V / 220V).	Unified horizontal button group with sliding active state.
quantity-selector	Incremental hardware packs.	Stepper (+ / -) interface with tiered volume discounts.
grid-cards	Multi-column machine comparisons.	2 or 3-column responsive card matrix highlighting model specs.

The 6 Selection Modes

SELECTION MODE	RULE & MATHEMATICAL CONSTRAINT	PRICING, STOCK & CART IMPACT
Single Selection	Customer chooses exactly 1 option per group.	Overrides base price. Validates stock for selected combination SKU.
Bundle Selection	Customer picks a package tier unlocking interactive slots.	Applies fixed bundle price or per-item rate. Validates stock across all chosen slot items.
Weight Based	Binds option directly to shipment gross weight.	Multiplies weight by quantity to calculate real-time courier shipping fees.
Multiple Selection	Customer can check multiple add-ons simultaneously.	Cumulatively sums all selected add-on prices to cart subtotal.
Quantity Based	Pack quantities with automatic volume discounting.	Decrements total pack count from central inventory.
Package Builder	Multi-step starter kit customizer.	Decrements multiple distinct warehouse SKUs upon single checkout completion.

COMBINATION MATRIX, PRICING & WEIGHT ENGINE

When multiple variant groups exist, clicking **Generate Combination Matrix** constructs a Cartesian permutation table.



Variant Weight & Shipping Formula

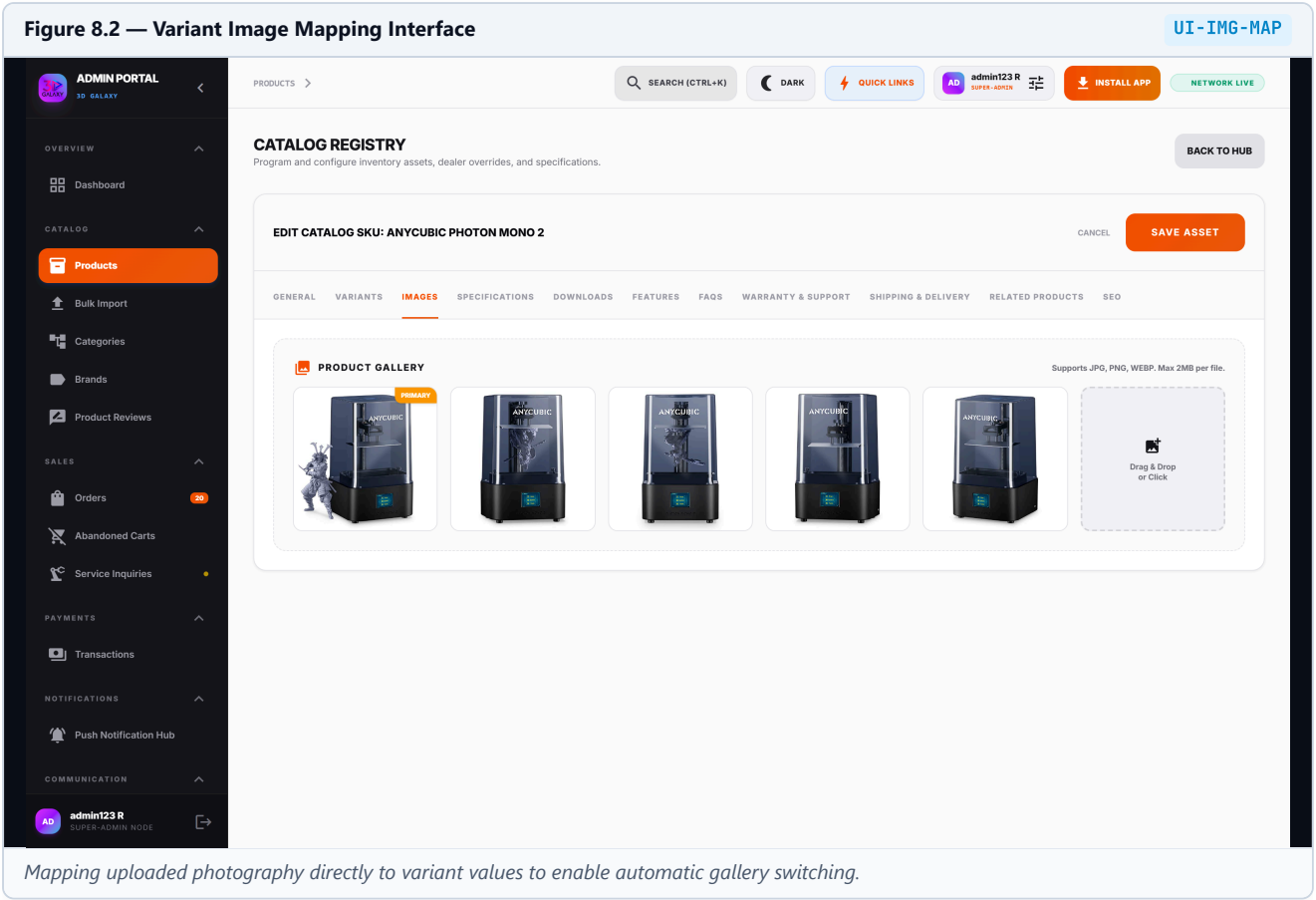
To avoid courier weight disputes and undercharged freight, variant weights are calibrated in grams:

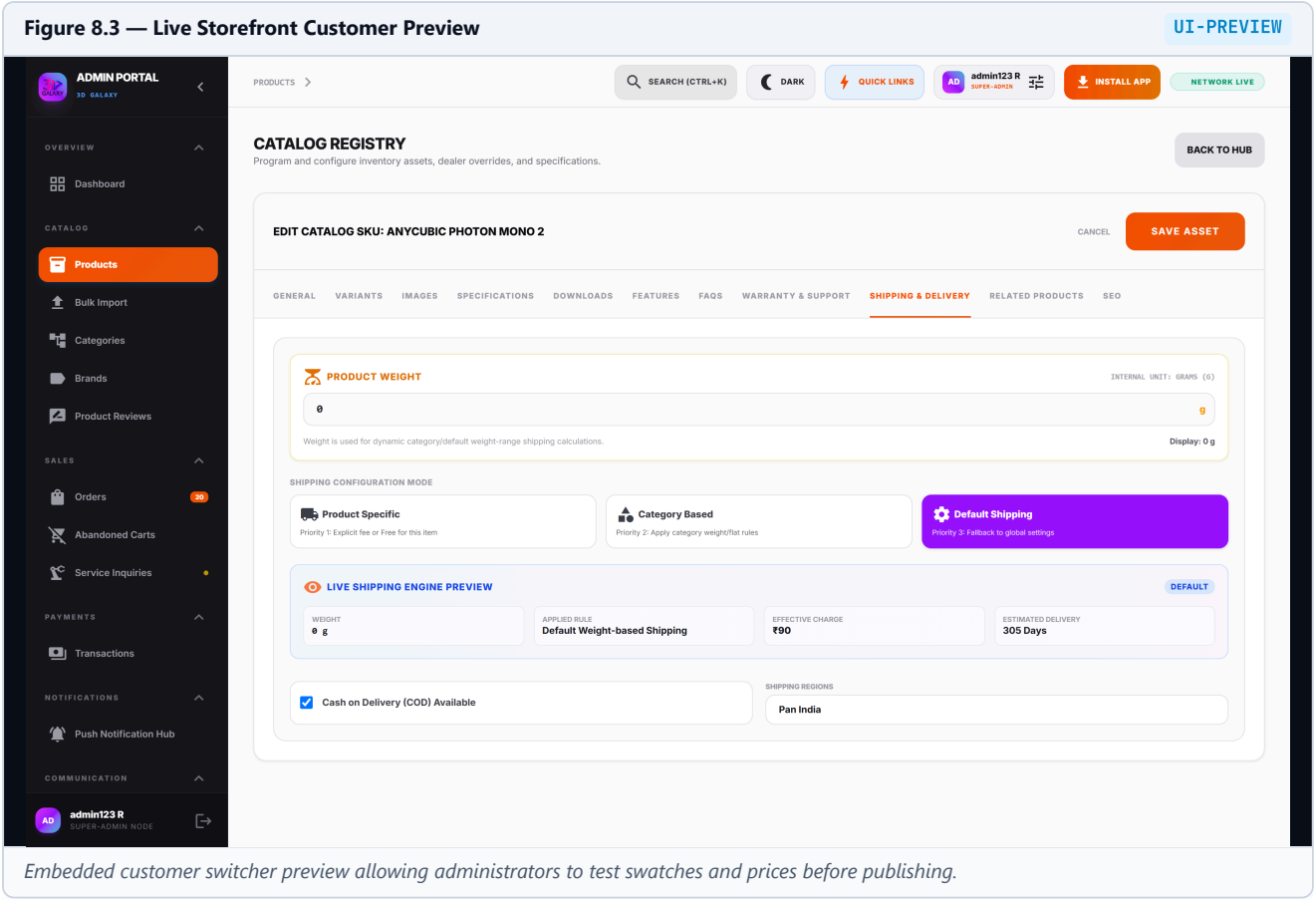
$$\text{Total Shipment Weight} = (\text{Variant Weight} \times \text{Quantity}) + \text{Tare Packaging Weight}$$

PRACTICAL WEIGHT CALCULATION EXAMPLE

When a customer purchases **4 spools of 250g sample filament**: $250\text{g} \times 4 = 1000\text{g} = 1.0\text{kg}$ The checkout automatically applies the **1.0kg courier tariff bracket**, ensuring accurate shipping recovery.

Variant Image Mapping & Storefront Preview





DYNAMIC VARIANT TEMPLATES & SAFETY RULES

The portal includes **10 pre-engineered templates** to eliminate repetitive manual configuration.

ADMIN PORTAL
3D GALAXY

OVERVIEW
Dashboard

CATALOG
Products

Bulk Import

Categories

Brands

Product Reviews

SALES
Orders 20

Abandoned Carts

Service Inquiries

PAYMENTS
Transactions

NOTIFICATIONS
Push Notification Hub

COMMUNICATION

admin123 R
SUPER-ADMIN

PRODUCTS

SEARCH (CTRL+K)

DARK

QUICK LINKS

admin123 R
SUPER-ADMIN

INSTALL APP

NETWORK LIVE

BACK TO HUB

CATALOG REGISTRY
Program and configure inventory assets, dealer overrides, and specifications.

EDIT CATALOG SKU: ANYCUBIC PHOTON MONO 2

CANCEL

SAVE ASSET

GENERAL VARIANTS IMAGES SPECIFICATIONS DOWNLOADS FEATURES FAQS WARRANTY & SUPPORT SHIPPING & DELIVERY RELATED PRODUCTS SEO

VARIANT SETUP GUIDE & ONBOARDING
Step-by-step interactive walkthrough for options, pricing, stock, images, and live previews.

TAKE VARIANT CONFIGURATION TOUR

Help

VARIANT TEMPLATES QUICK SETUP
Quickly configure common product variants (Filament Colors, Spool Weights, Bundles, Printers).

Hide Templates

All Templates Filament 3D Printer Accessories Bundles General

Search templates (e.g. Filament, Size, Bundle)...

Filament Color
FILAMENT
Single color selection using authentic 3D printing color chip pills.
Red Blue Green Yellow Orange +3 more
CHIP Preview & Use

Filament Weight
FILAMENT
Standard spool weight variants (250g, 500g, 1kg, 2kg, 5kg) with weight integration.
250g 500g 1kg 2kg 5kg
WEIGHT-SELECTOR Preview & Use

Material + Color
FILAMENT
Two-group matrix: Polymer Material Type x Vibrant Color Chips.
PLA PETG ABS TPU Resin
CHIP Preview & Use

Material + Weight + Color
FILAMENT
Comprehensive 3-group matrix: Material x Spool Weight x Color.
PLA PETG ABS
CHIP Preview & Use

Product Size
GENERAL
Physical apparel, enclosure, and 3D print sizing from XS to XXL.
XS S M L XL +3 more
CHIP Preview & Use

Storage / Capacity
ACCESSORIES
Digital storage & memory cards options (32GB up to 1TB).
32GB 64GB 128GB 256GB 512GB +1 more
CHIP Preview & Use

Predefined schemas with preview dialog, append mode protection, and collision warnings.

10 Verified Pre-Engineered Templates

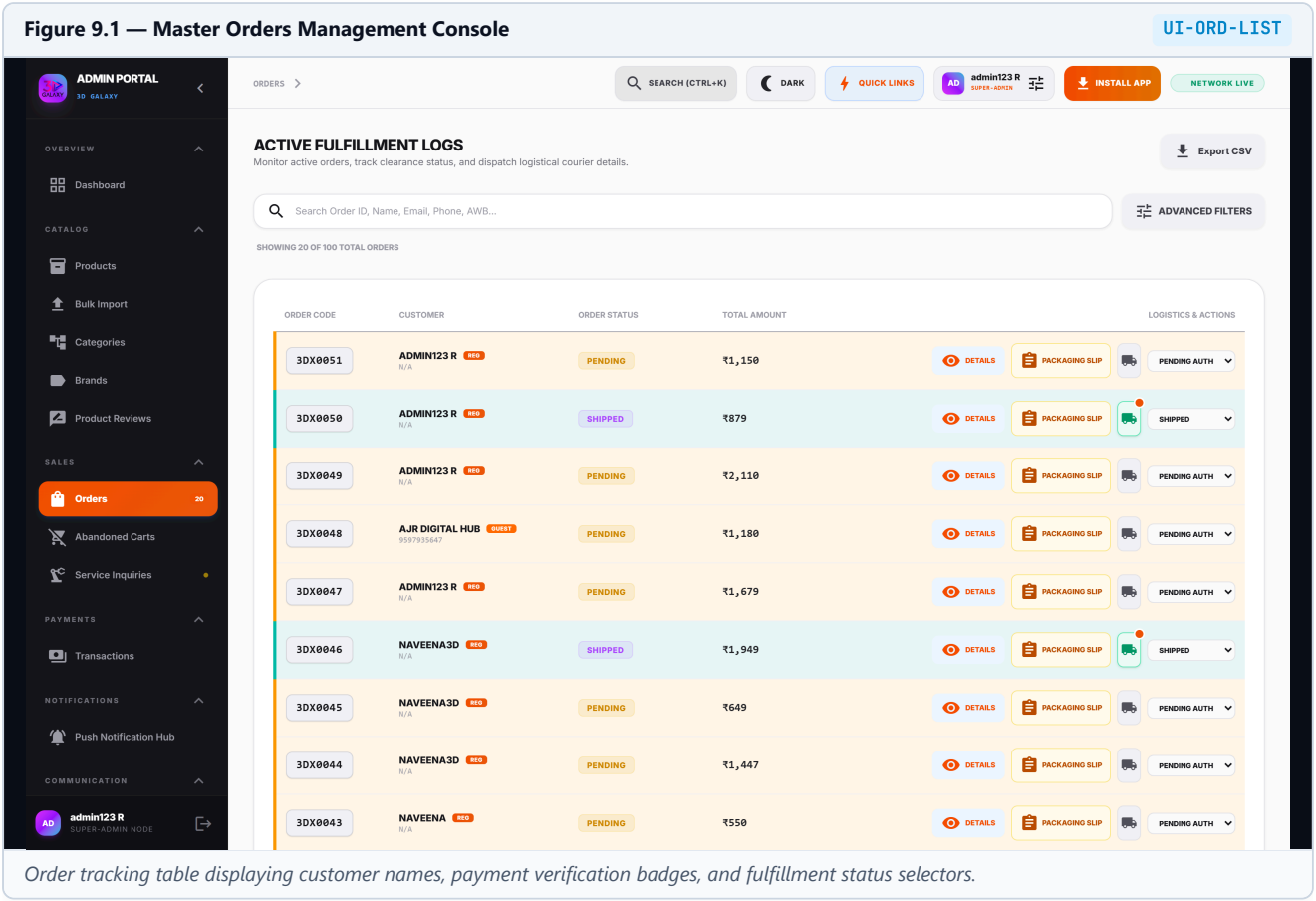
TEMPLATE NAME	CATEGORY	CONFIGURED SCHEMA & VALUES
Filament Color	Filament	1 Group: Color (Chip selector with 8 standard colors: Red, Blue, Green, Yellow, Orange, Black, White, Grey).
Filament Weight	Filament	1 Group: Spool Weight (Weight selector with 250g Sample, 500g, 1kg Standard, 2kg Value, 5kg Bulk).
Material + Color	Filament	2 Groups: Material (PLA, PETG, ABS, TPU, Resin) × Color (Color Chips: Red, Blue, Green, Black, White, Grey).
Material + Weight + Color	Filament	3 Groups: Material (PLA, PETG, ABS) × Weight (500g, 1kg) × Color (5 standard colors).
Product Size	General	1 Group: Size (XS, S, M, L, XL, XXL).
Storage / Capacity	Accessories	1 Group: Storage Capacity (32GB, 64GB, 128GB, 256GB, 512GB, 1TB).
Size + Color	Accessories	2 Groups: Size (S, M, L, XL) × Color (Red, Blue, Black, White).
Bundle Pack	Bundles	1 Group: Bundle Builder (Buy 1 @ ₹756, Buy 3 @ ₹642 each, Buy 5 @ ₹3,024 flat).
3D Printer Config	3D Printer	2 Groups: Printer Model (A1, A1 Mini, P1S, K1) × Package (Printer Only, Combo with AMS).
Custom Variant	General	Blank template ready for manual custom attributes.

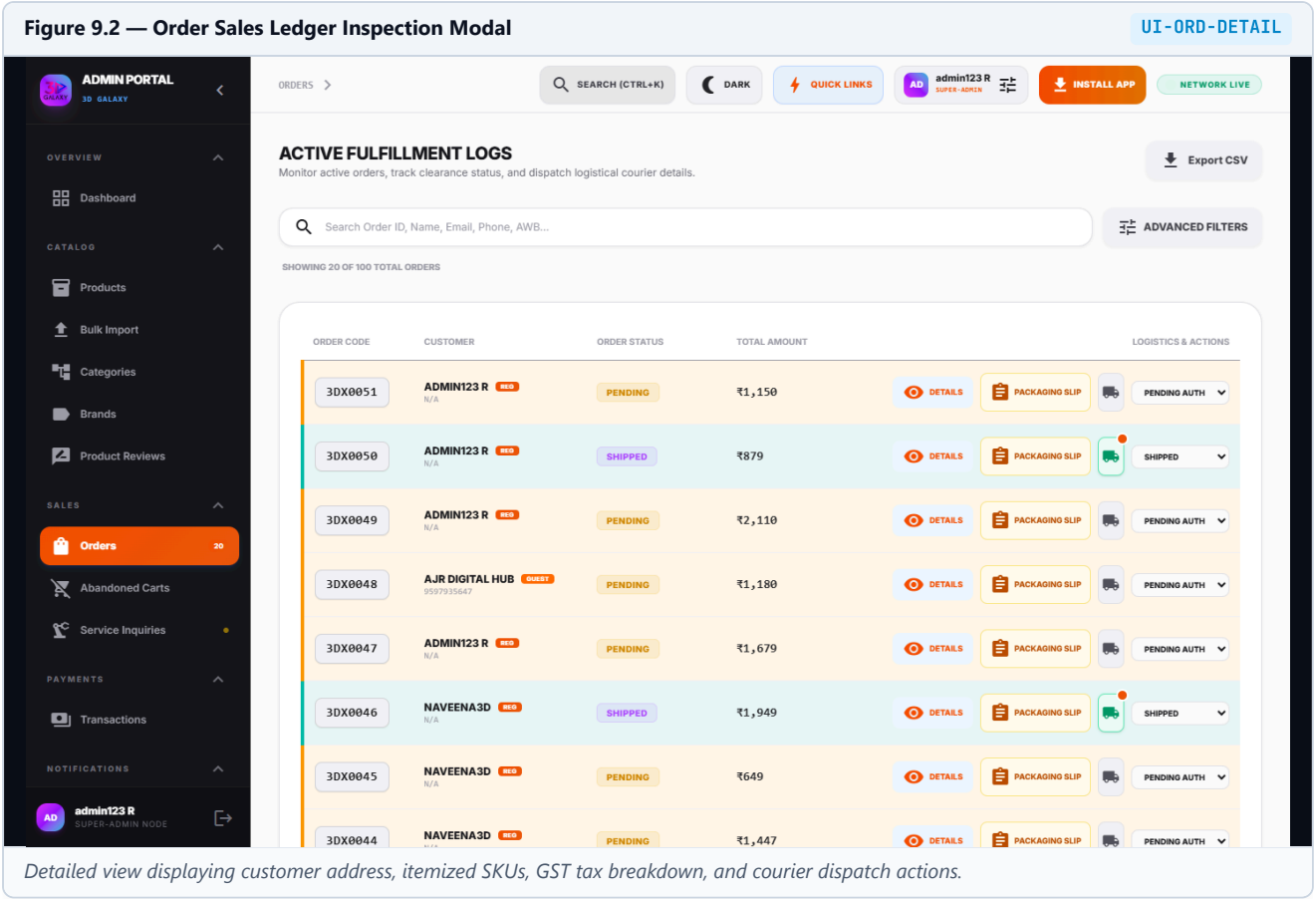
VARIANT TEMPLATE OPERATIONAL SAFETY RULES

- **No Silent Overwrites:** Template defaults to **APPEND** mode. Existing groups are preserved.
- **Collision Detection:** Warns if a group with the same name already exists on the product.
- **Template Independence:** Changing a template does not alter previously saved products.
- **Manual Stock Required:** Physical stock must always be entered manually based on actual inventory.
- **Explicit Image Binding:** Photos must be mapped under the *Images* tab after template application.

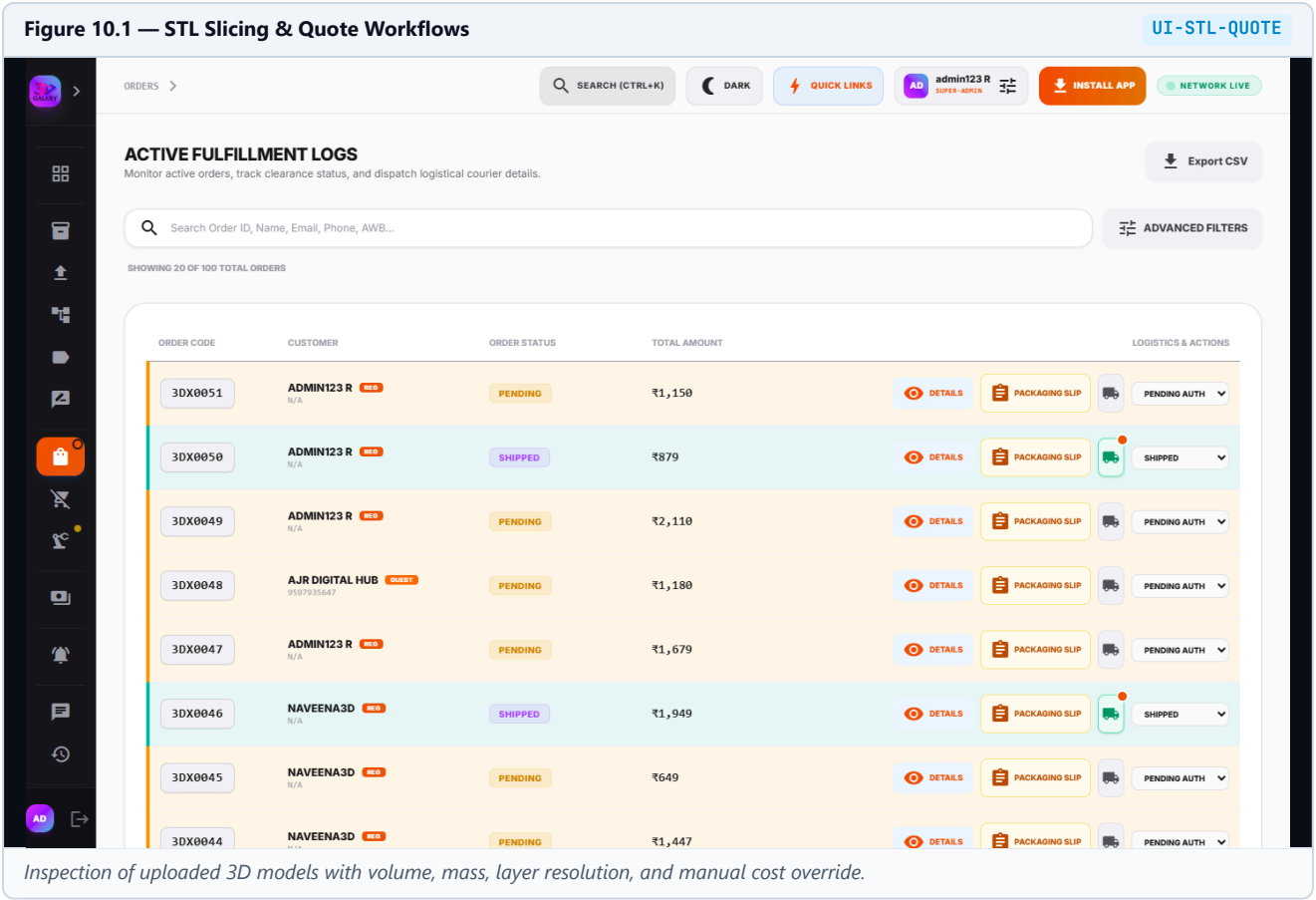
ORDERS, FULFILLMENT & STL SLICING QUOTES

The Orders console tracks client purchases from initial payment to delivery confirmation, including custom manufacturing orders.

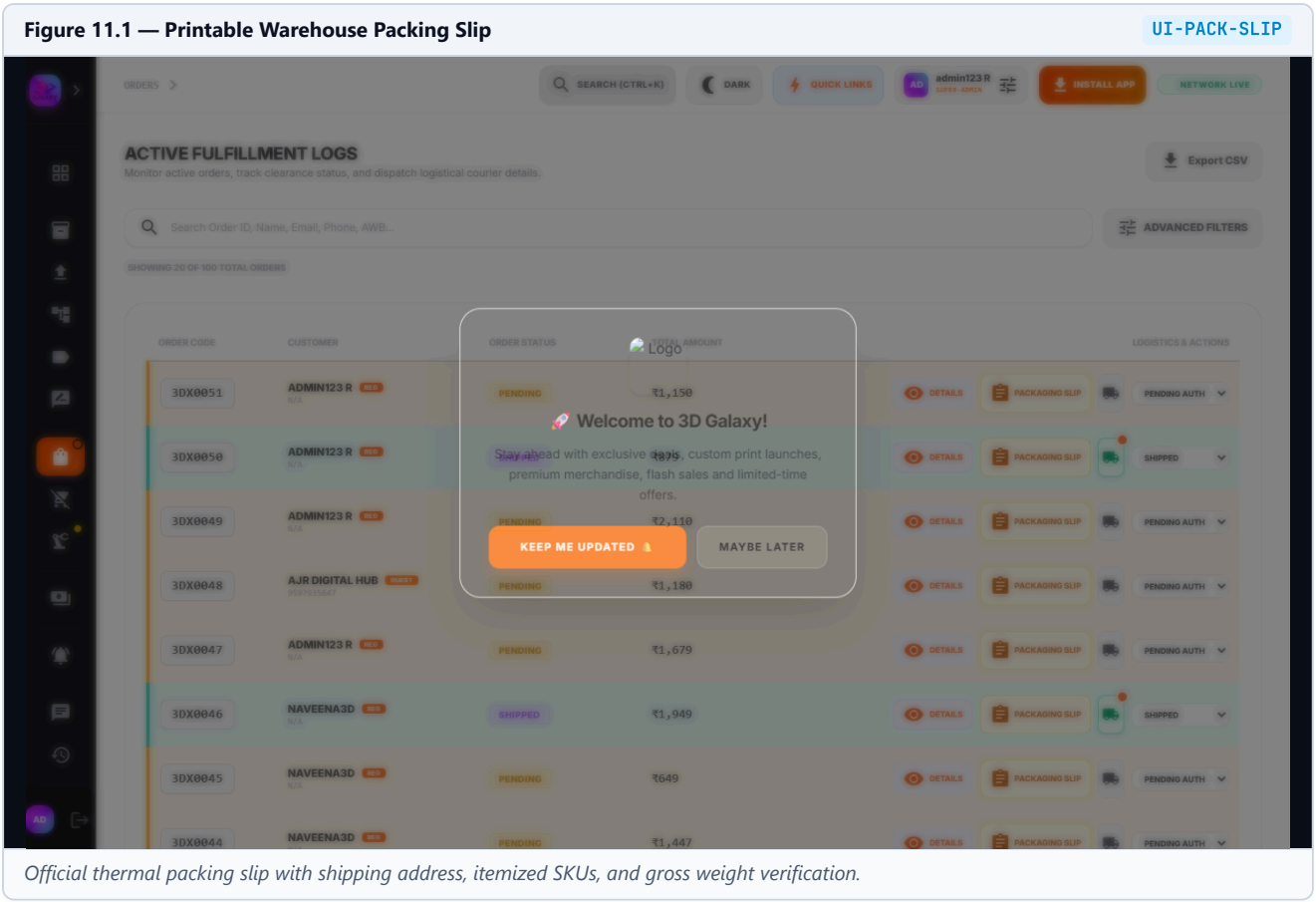




Customized 3D Model (STL) Slicing Quotes

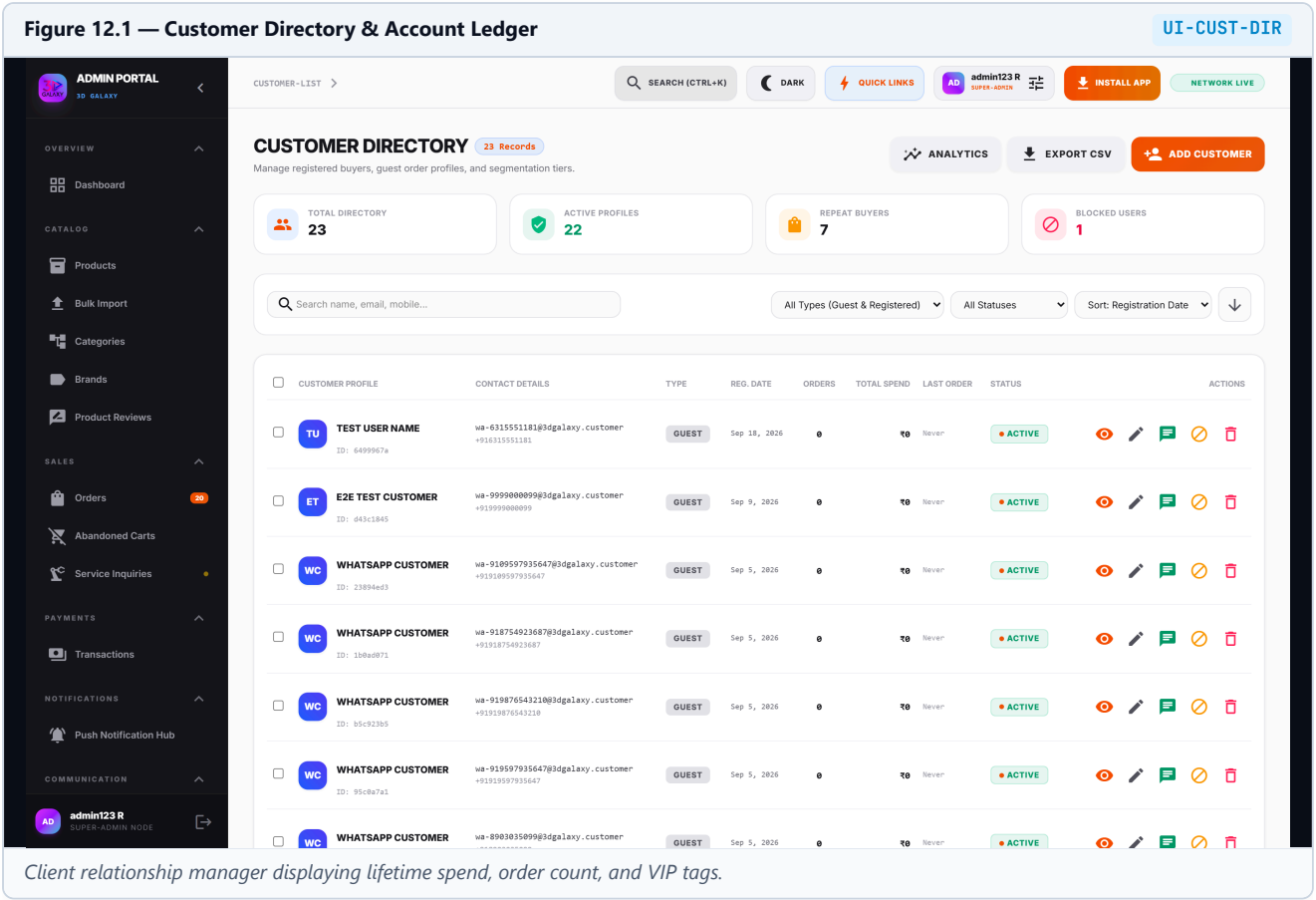


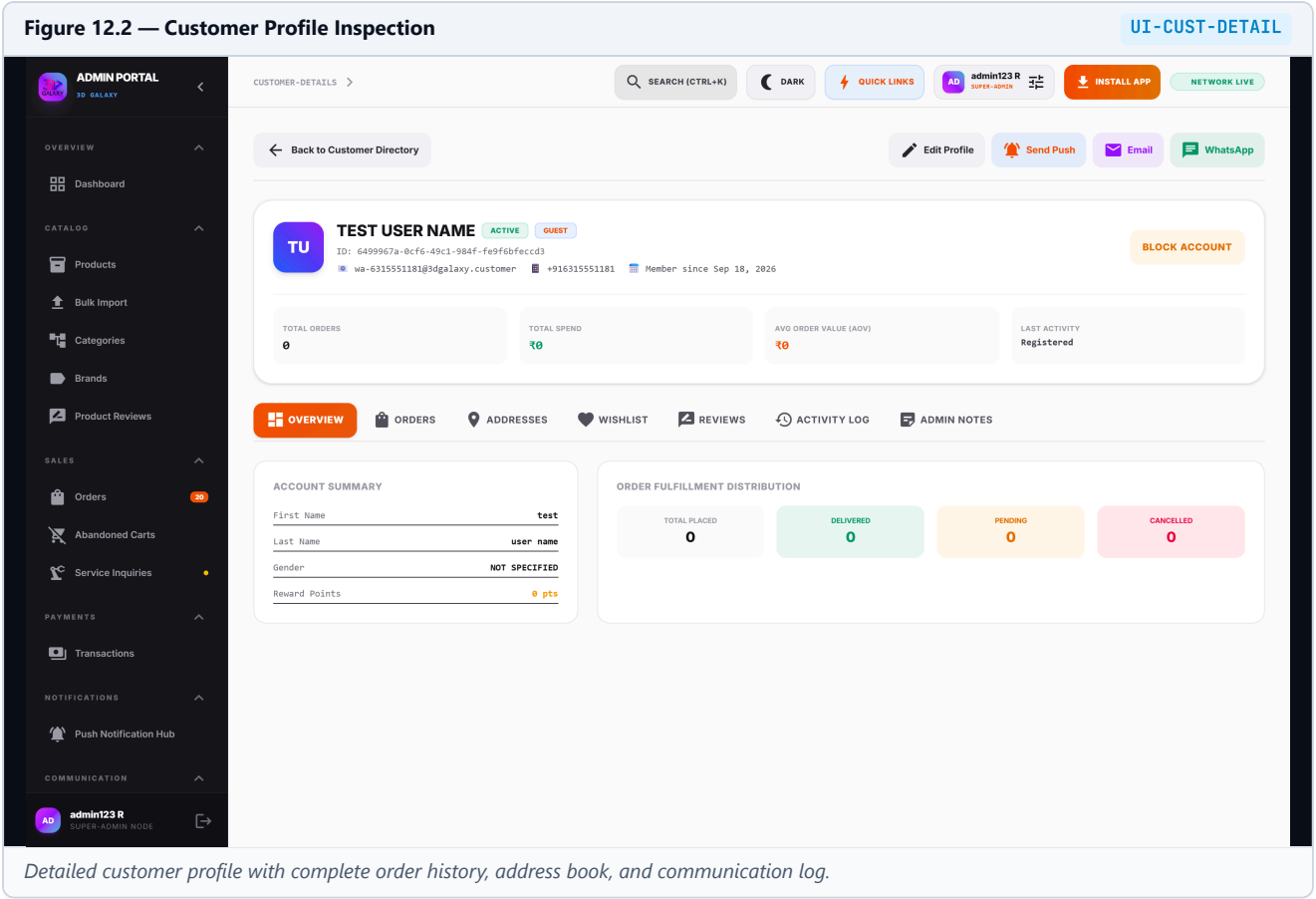
PACKING SLIPS, CUSTOMER DIRECTORY & ANALYTICS



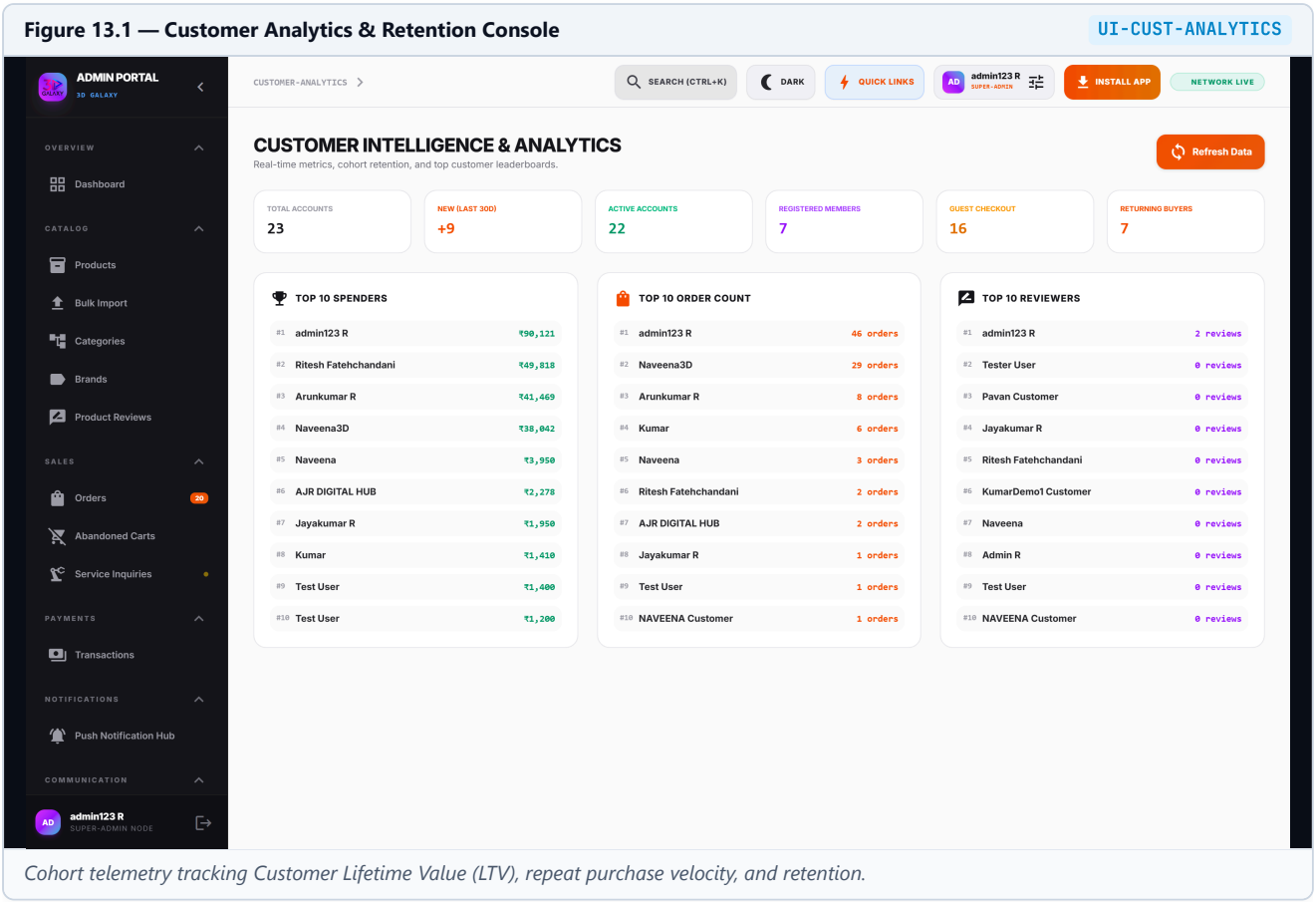
Official thermal packing slip with shipping address, itemized SKUs, and gross weight verification.

Customer Directory & CRM Dossier

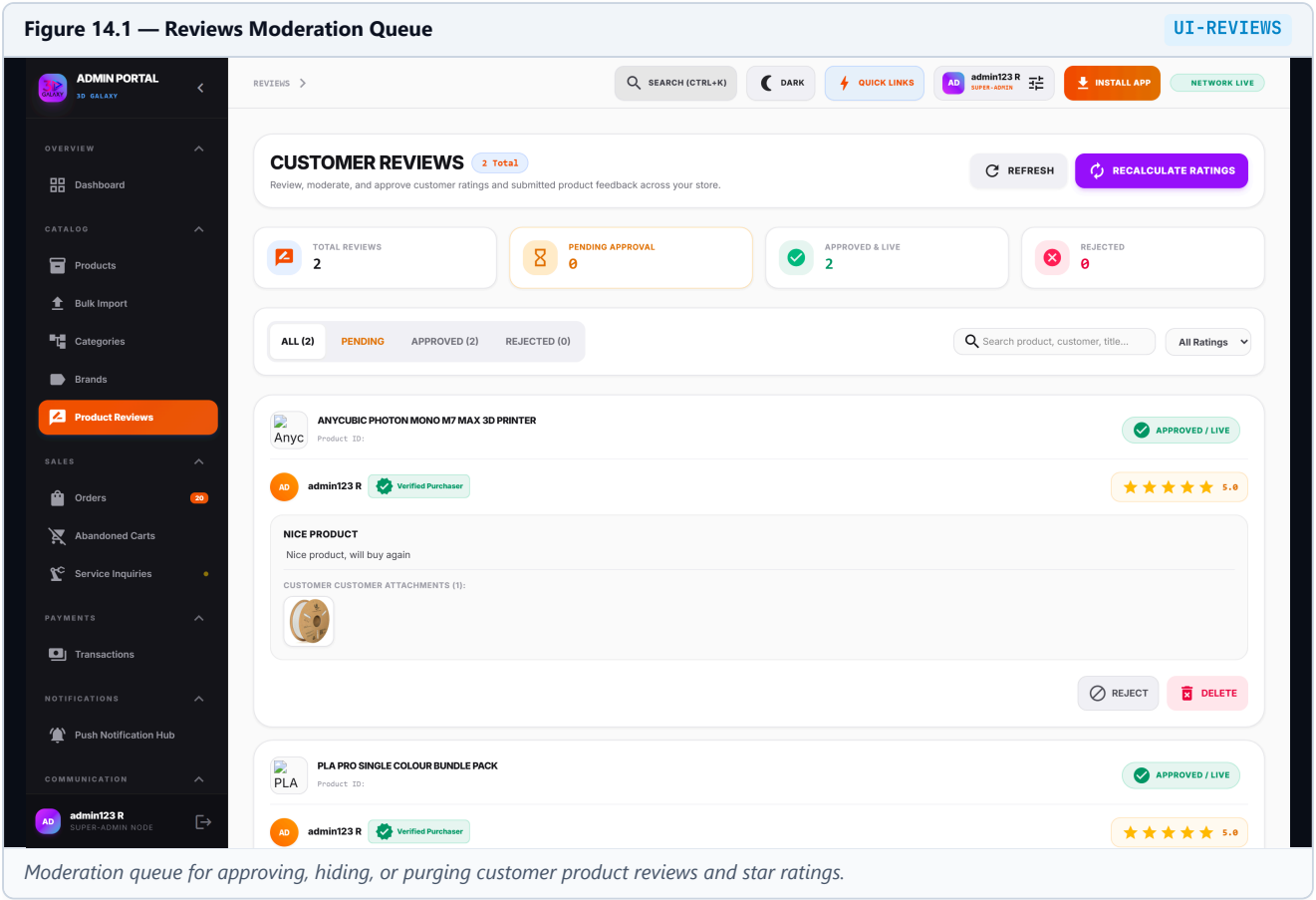




CUSTOMER ANALYTICS & REPUTATION MANAGEMENT

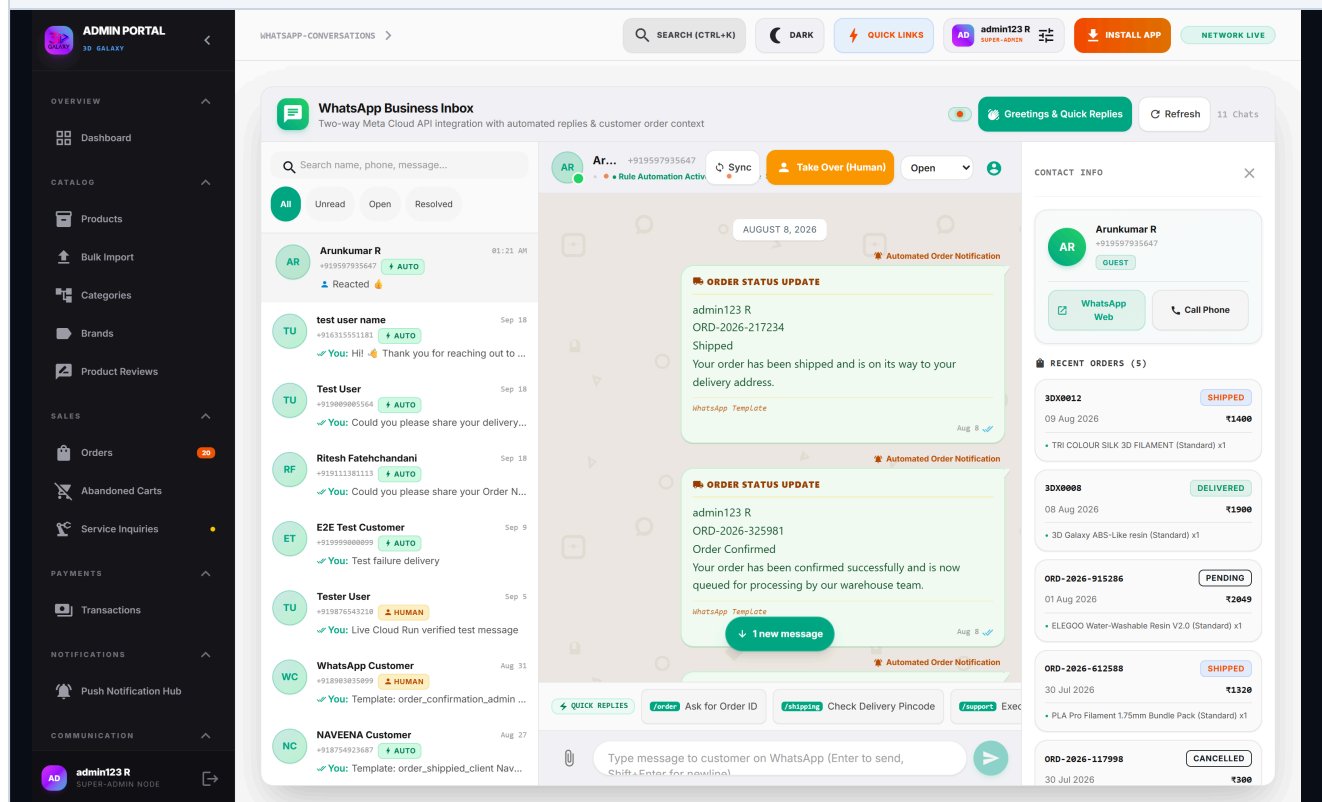


Customer Reviews Moderation Queue

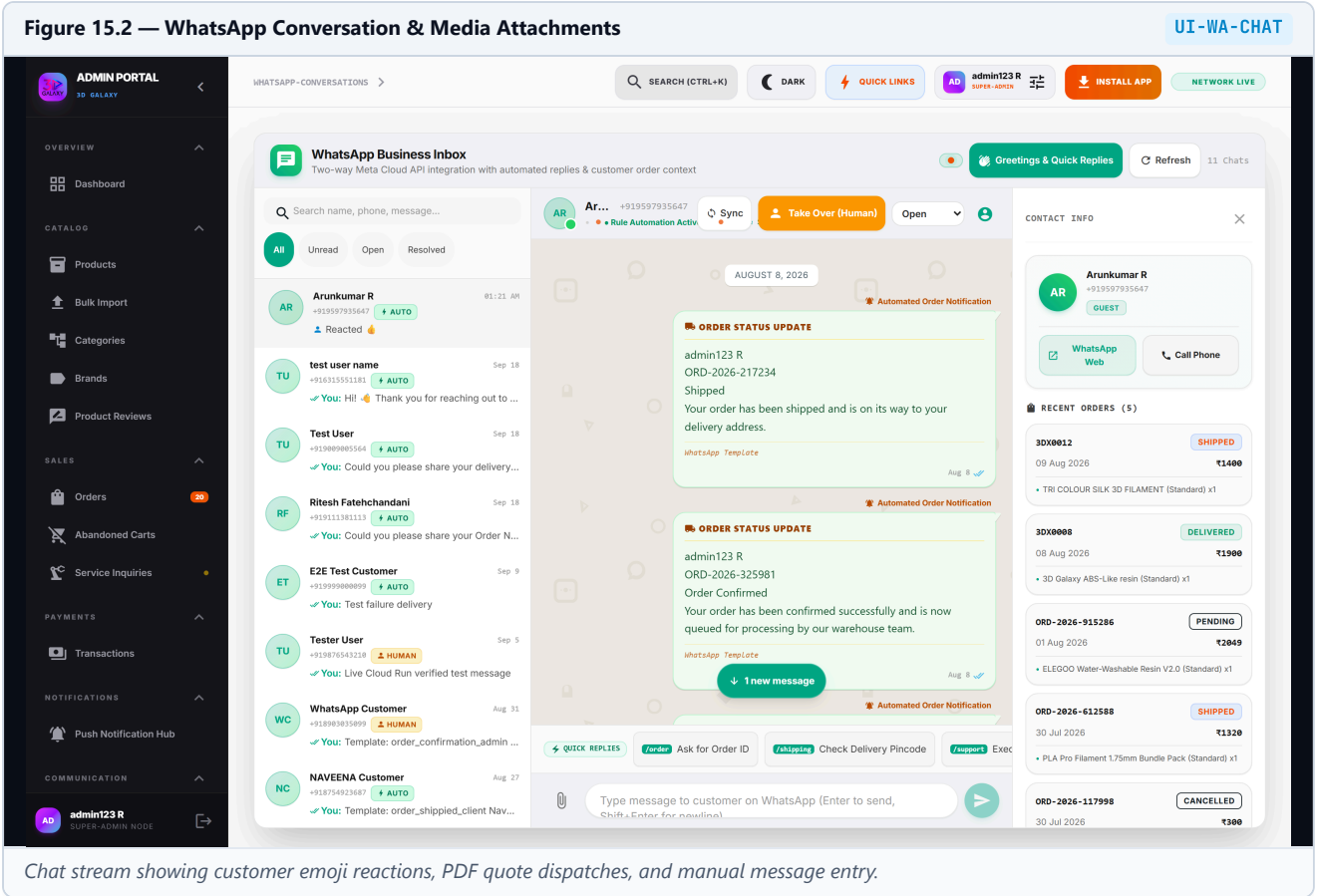


WHATSAAPP BUSINESS & PUSH NOTIFICATION HUB

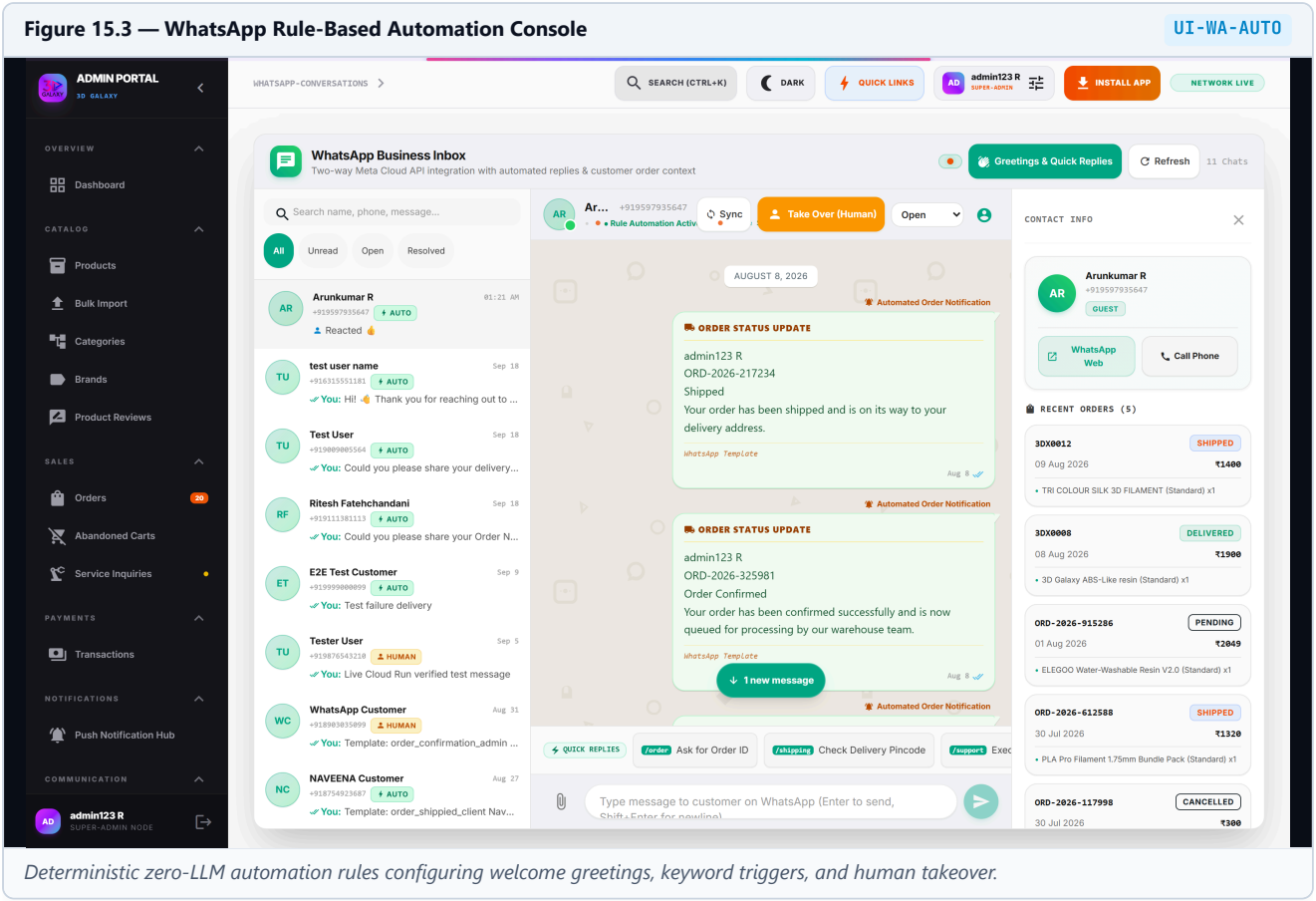
Figure 15.1 — WhatsApp Business Cloud API Live Inbox



Real-time dual-pane messaging inbox with unread badges, delivery ticks, and customer order context sidebar.



WhatsApp Rule-Based Automation Engine



Deterministic zero-LLM automation rules configuring welcome greetings, keyword triggers, and human takeover.

PUSH NOTIFICATIONS, MARKETING & CATEGORIES

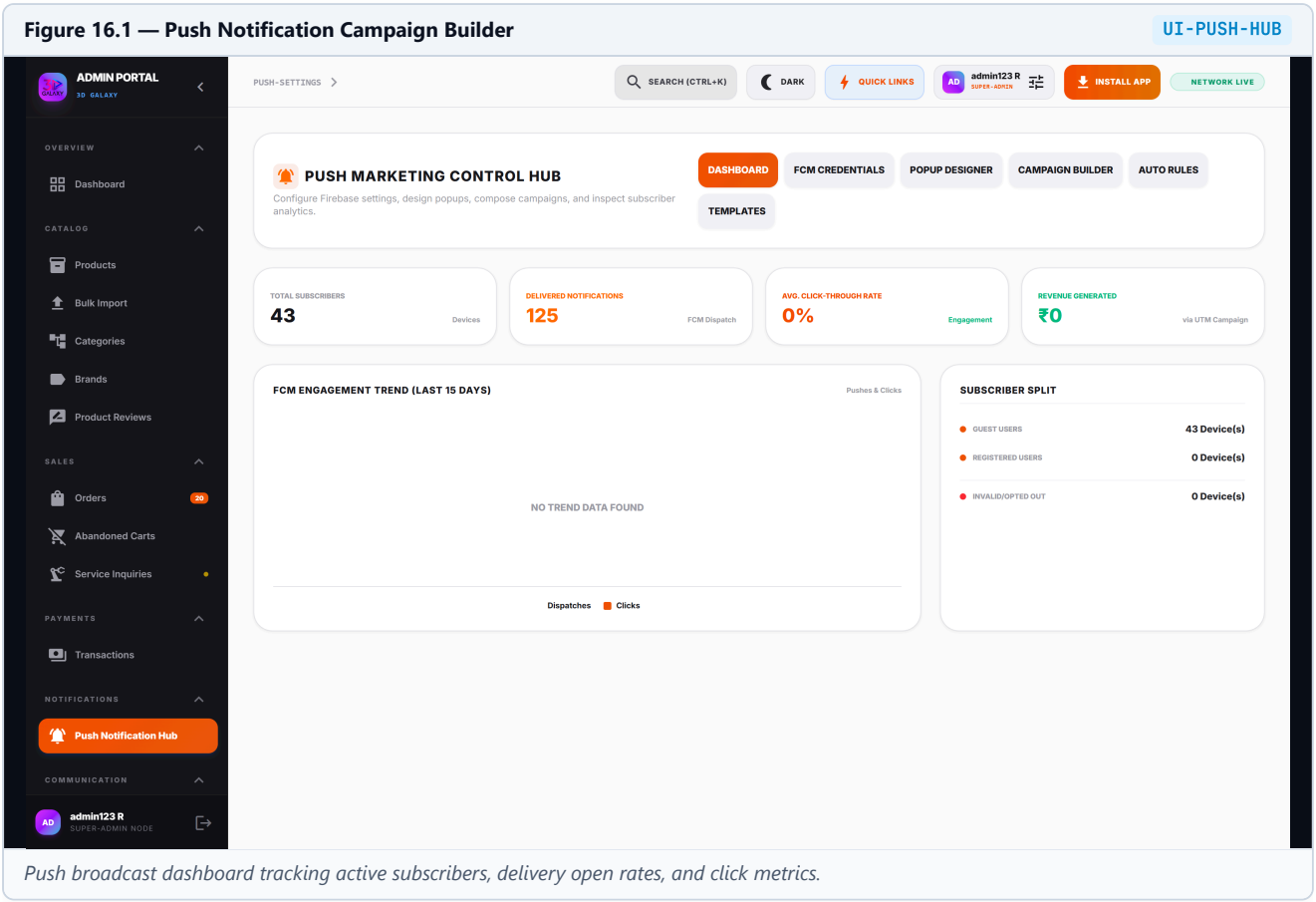
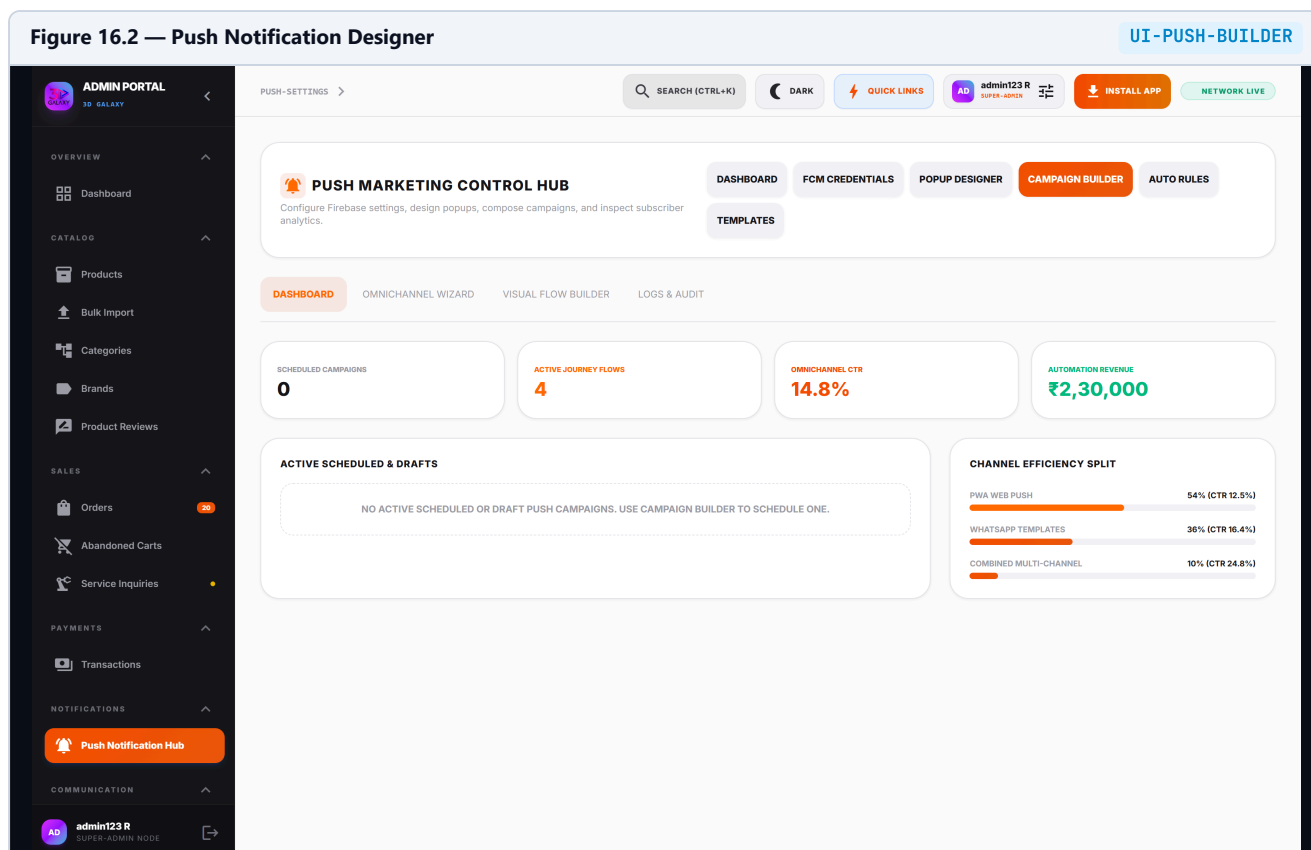
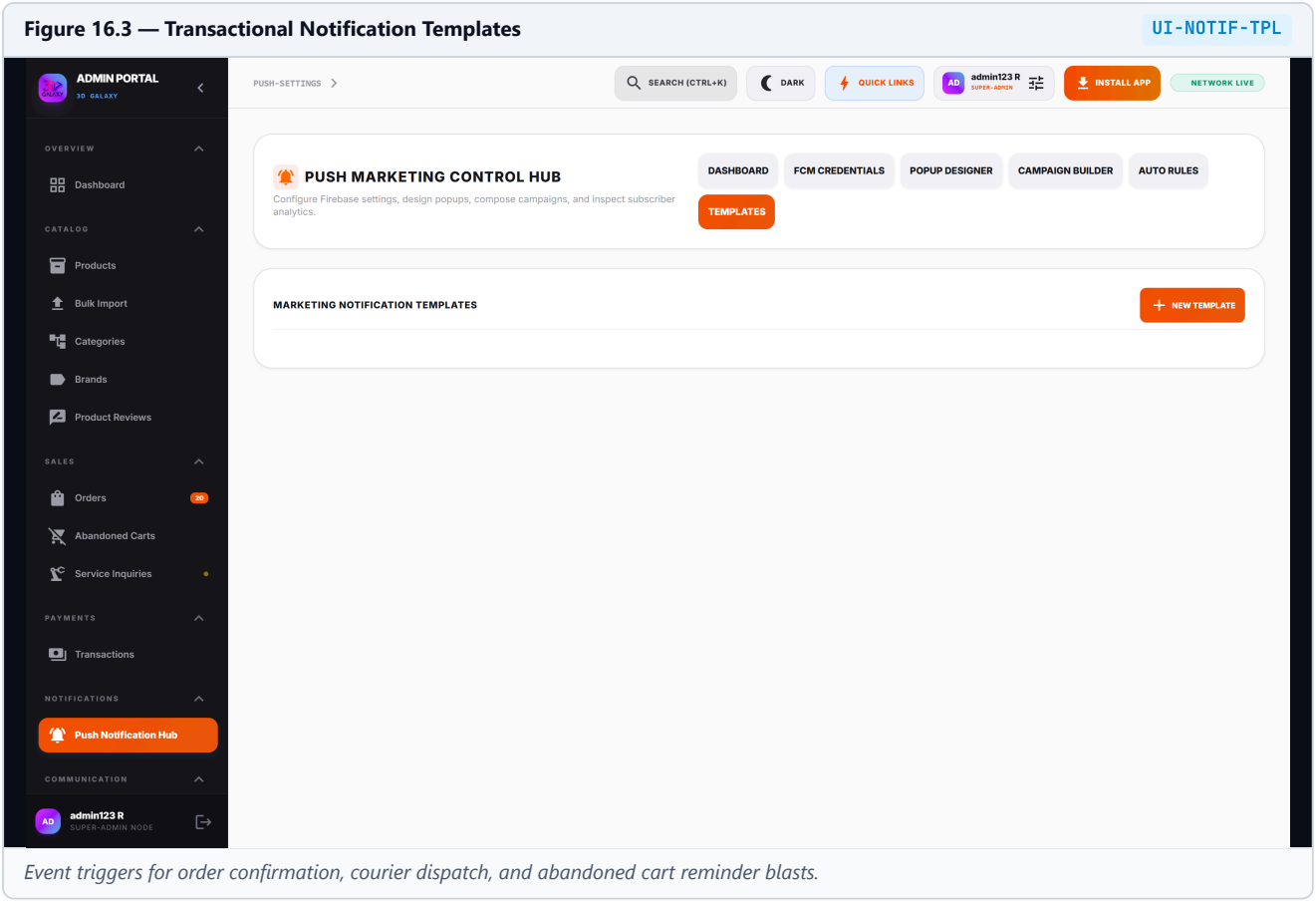


Figure 16.2 — Push Notification Designer

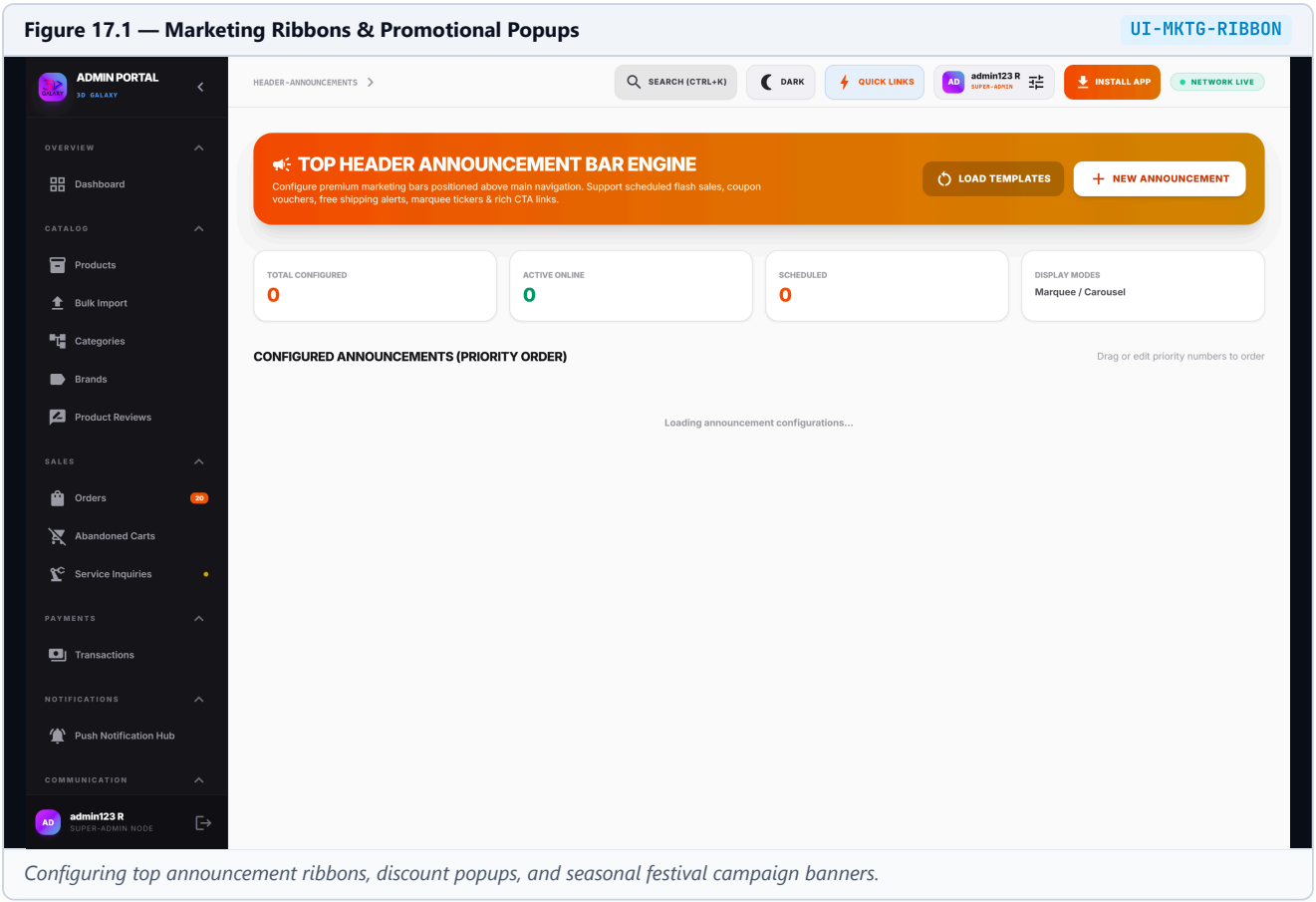


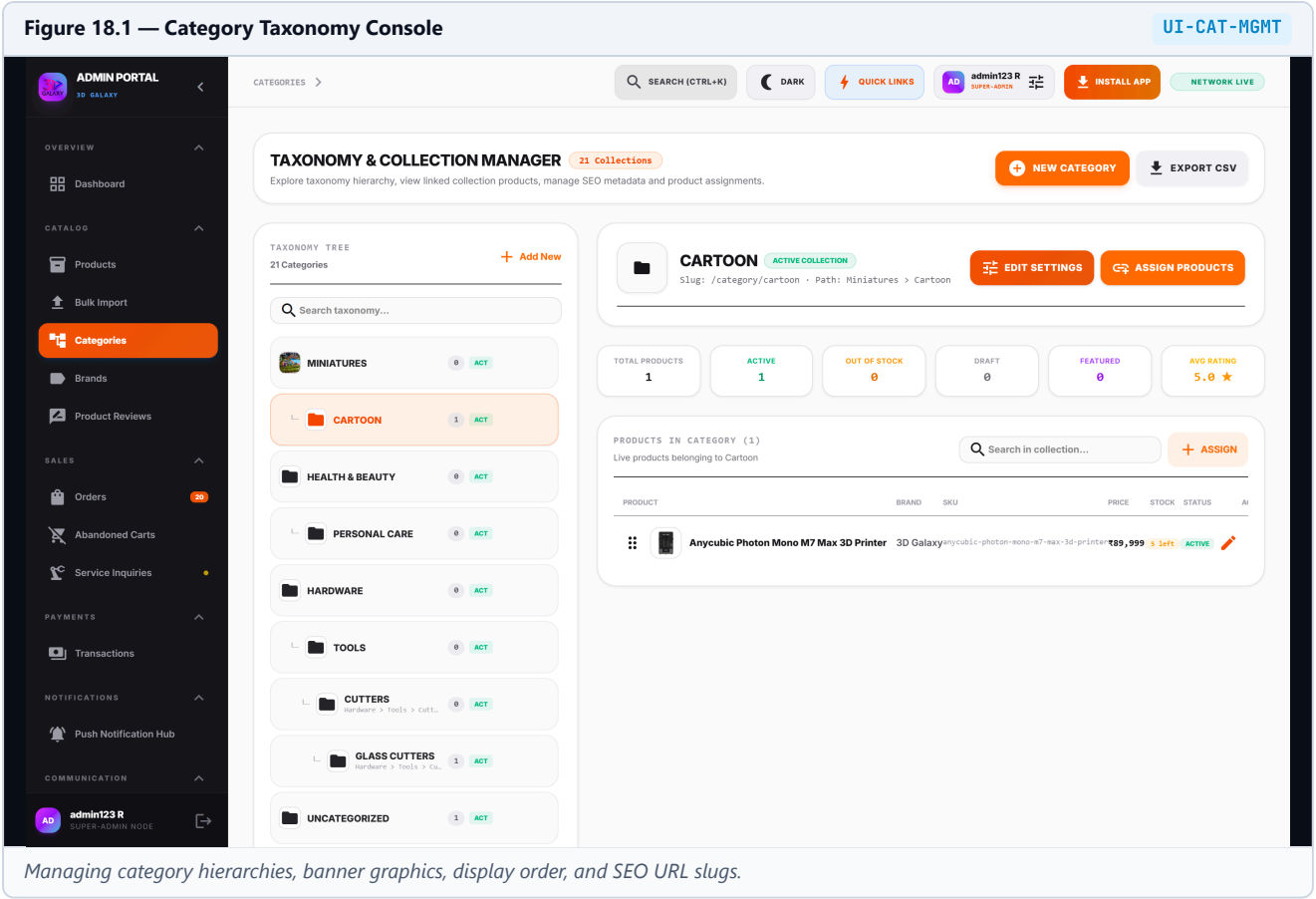
Campaign designer with audience segmentation, banner image preview, and deep-link routing.

Transactional Notification Templates & Triggers



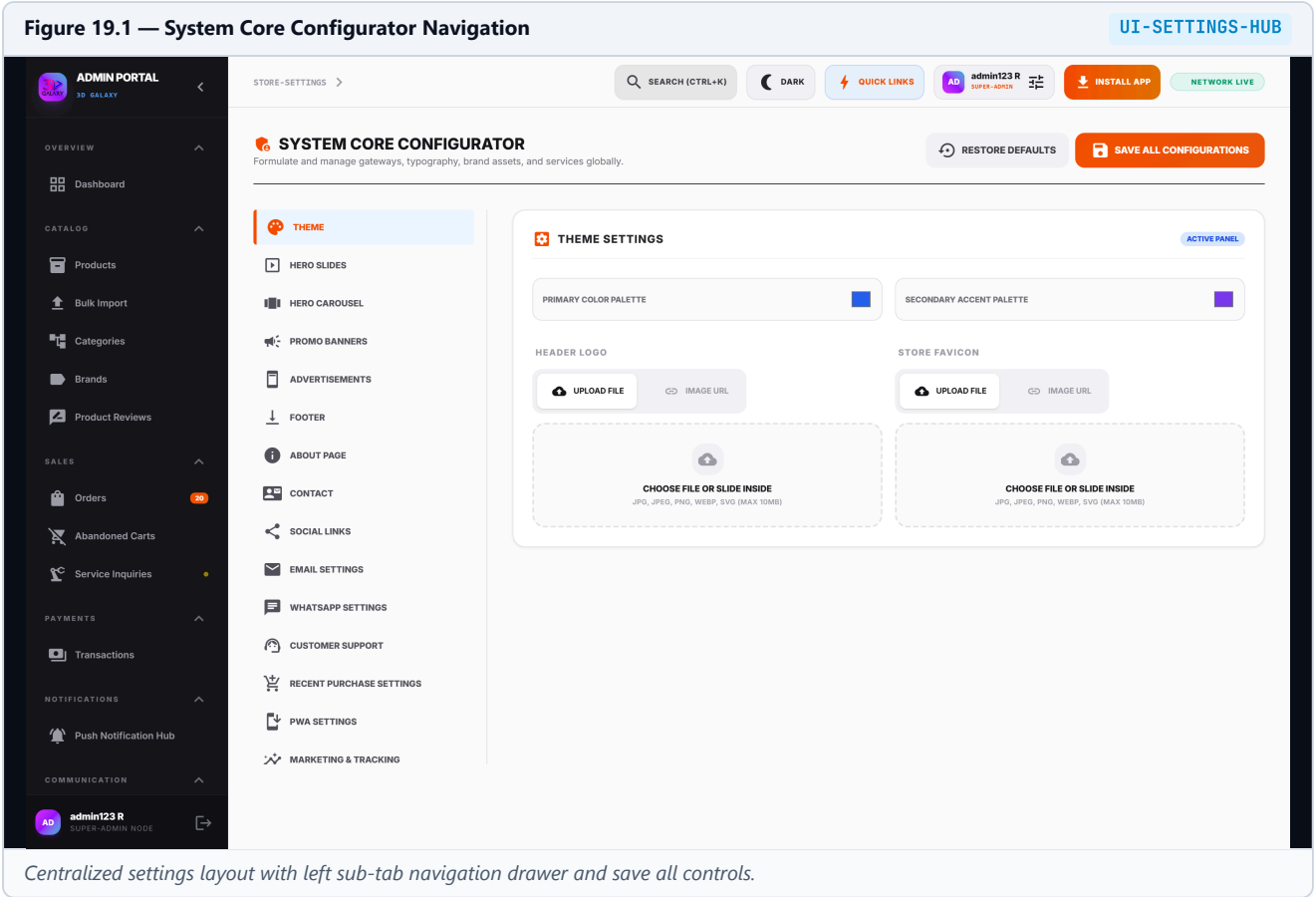
MARKETING BANNERS & CATEGORY TAXONOMY





SYSTEM CORE CONFIGURATOR (ALL 24 SUB-TABS)

The System Settings dashboard (‘/admin?tab=settings’) centralizes platform governance across **24 verified sub-tabs**.

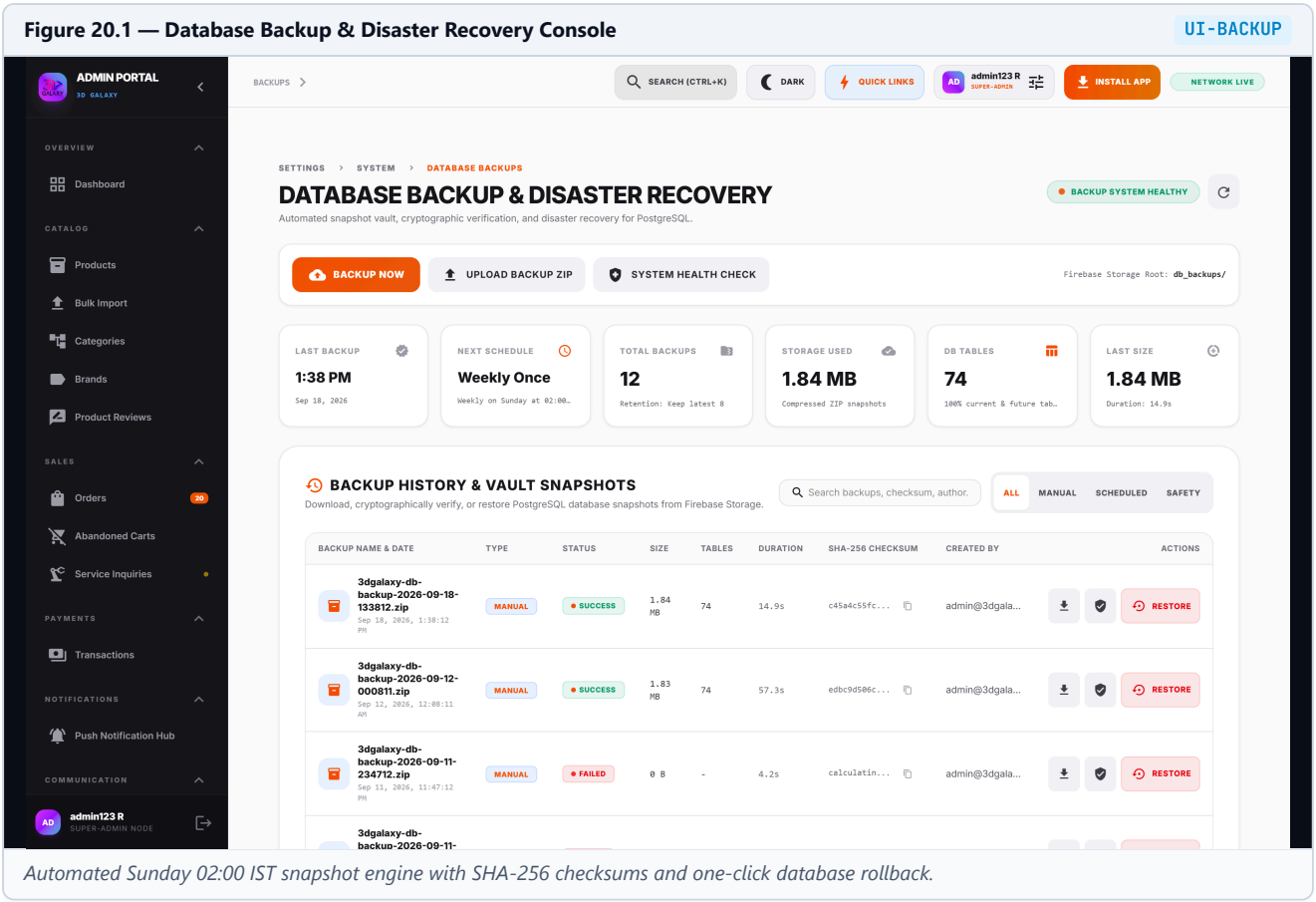


Comprehensive Breakdown of All 24 Settings Sub-Tabs

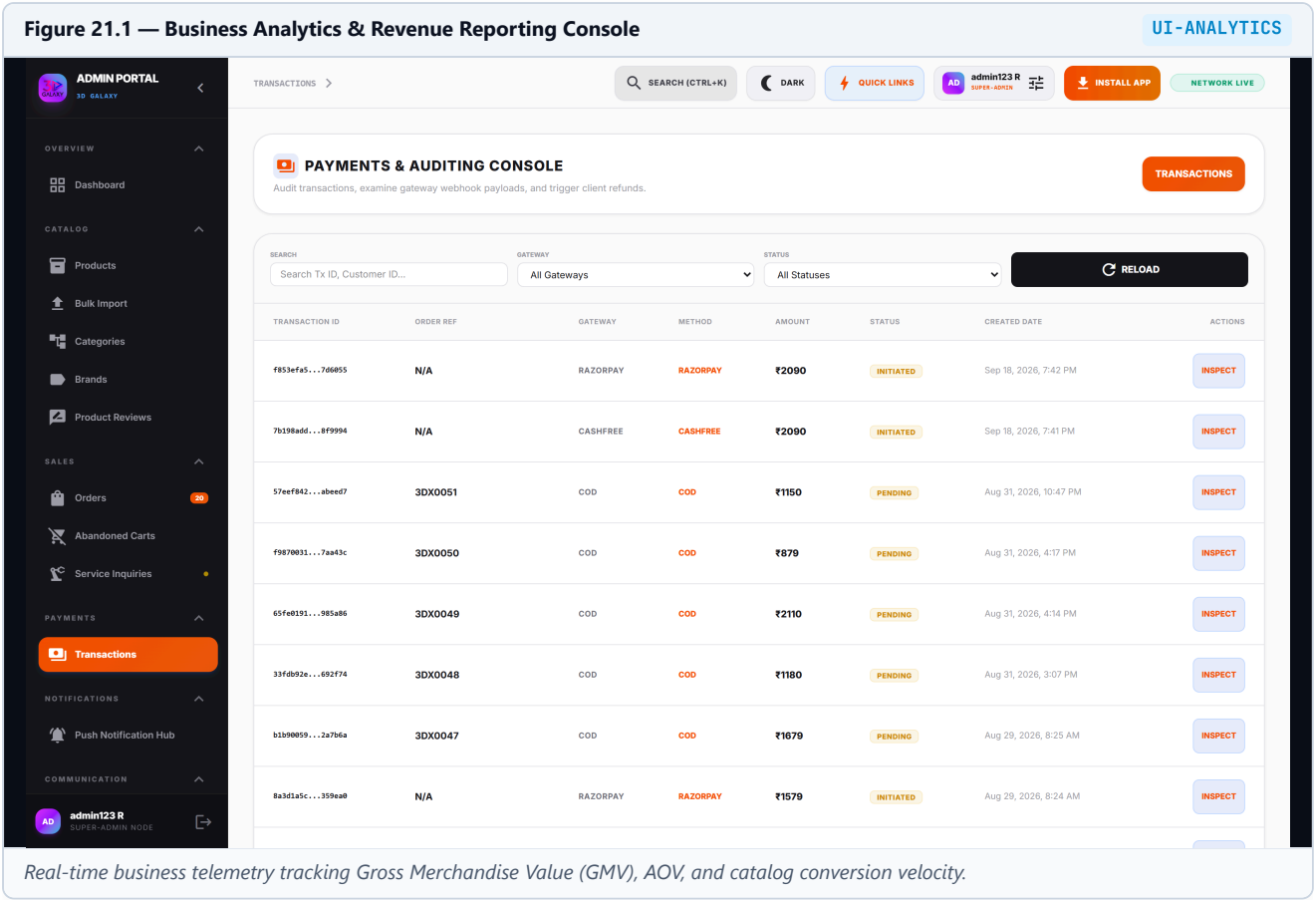
SETTINGS SUB-TAB	CONFIGURABLE SETTINGS & PARAMETERS	OPERATIONAL & BUSINESS IMPACT
1. Theme	Store Name, Tagline, Primary Color (#0284C7), Accent Color (#EA580C), Dark Mode.	Sets global brand styling, navigation colors, and default visitor theme.
2. Hero Slides	Slide Title, Subtitle, Desktop/Mobile Image URLs, CTA Link, Button Text, Duration.	Controls rotating hero carousel banners on homepage.
3. Hero Carousel	Autoplay toggle, slide interval (ms), animation physics (fade/slide), touch swipe.	Governs animation speed and responsive gesture behavior.
4. Promo Banners	Promo Headline, Discount Text Badge, Image Asset URL, Deep Link.	Highlights seasonal sales between product rows on homepage.
5. Advertisements	Partner Ad ID, Banner Graphic, Destination URL, Display Zone.	Monetizes commercial partner promotional tiles.
6. Footer	Copyright Text, Registered Address, GSTIN, Customer Care Hotline, Payment Badges.	Establishes legal credibility and footer navigation links.
7. About Page	Mission Statement, Lab Facility Photos, Engineering Bio, Quality Statement.	Builds buyer confidence in 3D Galaxy's manufacturing capabilities.

8. Contact	Support Email, Emergency Phone, Warehouse Address, Google Maps Iframe Embed.	Directs client inquiries to official operational hotlines.
9. Social Links	Instagram, YouTube, Facebook, LinkedIn, X/Twitter, Discord, GitHub URLs.	Connects buyers to community build logs and video tutorials.
10. Email Settings	SMTP Host, Port (587/465), Sender Email, Display Name, Auth Tokens *(Masked)*.	Delivers order confirmations and invoice receipts.
11. WhatsApp Settings	Phone Number ID, WABA ID, Graph API Version (v20.0), Webhook Challenge Token.	Authenticates bidirectional WhatsApp Cloud API messaging.
12. Customer Support	Support Channels (Live Chat, WhatsApp, Phone), SLA Response Time Display.	Sets buyer expectations for support resolution windows.
13. Recent Purchases	Social proof popups, display interval (seconds), toast duration, product pool.	Creates buyer urgency and social validation on storefront.
14. PWA Settings	Enable PWA Module, Caching Strategy, Offline Fallback, Auto-Update Prompt.	Allows mobile administrators to install the portal as a native web app.
15. Marketing & Tracking	Meta Pixel, Meta CAPI Token *(Masked)*, GA4 Stream ID, Google Ads Conv ID, GTM.	Powers omnichannel conversion attribution and remarketing.
16. Admin Devices	Registered Devices List, OS/Browser Telemetry, FCM Tokens, Revoke Access.	Secures administrative accounts and tracks active terminal sessions.
17. Shipping	Default Fee (₹99), Free Shipping Threshold (₹1,999), Weight Rules Table, Couriers.	Automates shipping charges and tracking link generation.
18. Payment Gateway	Razorpay Key ID/Secret *(Masked)*, Cashfree, COD Enable, Max COD Threshold.	Processes real-time payments and sets fraud limits.
19. Newsletter	Opt-in Popup Toggle, Welcome Voucher Code (WELCOME3D), Export Subscribers.	Captures prospective client leads for promotional email blasts.
20. Product Page	Sticky Mobile Buy Bar, Low Stock Threshold, Delivery Calculator Display.	Maximizes mobile conversion rate and displays shipping countdowns.
21. FAQ	Global Accordion Items (Category, Question, Rich Text Answer, Sort Order).	Answers common storewide shipping and payment questions.
22. Services	Commercial Service Tiles (3D Scanning, Slicing Consulting, Prototyping).	Promotes B2B industrial additive manufacturing services.
23. 3D Printing Service	Materials (PLA, PETG, ABS, Resin, TPU), Prices/g, Densities, Layer Heights, Infills.	Powers automated real-time custom STL slicing quote calculations.
24. Database Backups	Manual Snapshot, Automated Sunday Cron, Storage Path (db_backups/), Restore.	Guarantees zero data loss with cryptographic SHA-256 validation.

DISASTER RECOVERY & BUSINESS ANALYTICS



Business Analytics & Revenue Telemetry



20 PRACTICAL ADMINISTRATOR OPERATIONAL USE CASES

#	OPERATIONAL USE CASE	STEP-BY-STEP EXECUTION PROCEDURE
01	Create a New Product	Open Products $\rightarrow$ Add Product. Enter Title, SKU, Category, MRP, Sale Price, Stock, upload photo, and click Publish.
02	Configure Categories	Open Categories $\rightarrow$ Add Category. Enter name, slug, upload banner asset, assign parent category, and save.
03	Configure Single Variant	Under Variants tab, add Group "Color" (<code>color-chips</code>), add values with hex codes, generate matrix, set stock, and save.
04	Configure Multiple Variants	Create Group 1 (Material), Group 2 (Weight), Group 3 (Color). Generate Combination Matrix, set stock per row, and save.
05	Apply Variant Template	Click Variant Templates Library $\rightarrow$ select "Filament Weight" $\rightarrow$ click Apply. Adjust prices and allocate stock.
06	Configure Variant Stock	In Combination Matrix table, locate target permutation row, enter warehouse stock counter, and click Save Product.
07	Configure Variant Weight	Enter calibrated weight in grams in the weight column (e.g. 1kg spool = 1250g with packaging) to calibrate courier tariffs.
08	Link Variant Images	Under Images tab, upload color photos. From "Linked Variant" dropdown on each photo, select corresponding color value.
09	Configure Bundle Pack	Set Display Type to <code>bundle-builder</code> , configure tiers (Buy 1, Buy 3, Buy 5), set pricing type and savings badges.
10	Configure Shipping Rules	In product Shipping tab, set Delivery Days buffer to 4 days, toggle Free Shipping if promotional, and save.
11	Process Active Order	Open Orders $\rightarrow$ Active Logs. Verify payment ID, change status from <code>pending</code> to <code>processing</code> , and print packing slip.
12	Process Customized STL Job	Open STL Slicer Quotes $\rightarrow$ inspect volume (cm^3) and mass (g). Enter manual cost override and click Dispatch Quote.
13	Ship Order & Tracking	Change status to <code>shipped</code> , select Courier (e.g. Delhivery), paste tracking number, and click Save to alert customer.
14	Update Tracking Number	Open order modal, correct the consignment number, and save. The customer's tracking link reflects the new number immediately.
15	Handle WhatsApp Inquiries	Open WhatsApp Business $\rightarrow$ select conversation thread. Review customer order history in sidebar and type reply.
16	Configure WhatsApp Rules	Open WhatsApp Automation $\rightarrow$ enable Welcome Greeting $\rightarrow$ add keyword rules (e.g. "FILAMENT" returns catalog link).
17	Launch Push Campaign	Open Push Notifications $\rightarrow$ Campaign Builder. Input title, body, banner image, deep-link URL, and click Send Broadcast.
18	Audit Business Analytics	Open Analytics $\rightarrow$ review GMV, AOV, and Low Stock tickers. Issue replenishment orders for items with < 5 units.
19	Create Manual Backup	Open Settings $\rightarrow$ Database Backups. Verify system status is Operational $\rightarrow$ click Create Manual Snapshot.

20	Restore Database Snapshot	In Database Backups, locate desired snapshot by timestamp \$\\rightarrow\$ click Restore \$\\rightarrow\$ confirm critical prompt.
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OPERATIONAL WORKFLOWS & CLIENT CHECKLISTS

End-to-End Product Workflow

```
CREATE PRODUCT → GENERAL DETAILS → CATEGORIES → IMAGES → VARIANTS / TEMPLATES
    ↓
COMBINATION MATRIX → PRICING & STOCK → WEIGHT CALIBRATION → IMAGE MAPPING
    ↓
SHIPPING RULES & SEO → LIVE PREVIEW VALIDATION → PUBLISH → STOREFRONT CART & CHECKOUT
```

End-to-End Order Workflow

```
CUSTOMER CART → CHECKOUT & PAYMENT → PENDING ORDER → ADMIN VERIFICATION
    ↓
PROCESSING → WAREHOUSE PICKING → PACKING SLIP → COURIER DISPATCH & TRACKING
    ↓
AUTOMATED WHATSAPP / EMAIL NOTIFICATION → OUT FOR DELIVERY → COMPLETED FULFILLMENT
```

Pre-Publishing Checklist (Products)

- Product Title is canonical with brand and model
- SKU is unique and matches warehouse bins
- Primary Category and secondary tags assigned
- MRP and Retail Sale Price verified
- Calibrated weight entered in grams
- High-resolution photos uploaded (min 1000×1000px)
- Variant groups and Combination Matrix generated
- Default variant star placed on in-stock option
- Variant photos linked under Images tab
- Delivery estimate buffer days configured
- SEO title and description tags populated
- Validated in Live Customer Preview switcher

Pre-Fulfillment Checklist (Orders)

- Payment verified (Razorpay ID valid or COD authorized)
- Customer phone number and shipping address verified
- GSTIN verified for corporate tax credit invoice
- All ordered SKUs and variant options in stock
- Custom STL slicing specs reviewed (if manufacturing)
- Shipping label printed with accurate parcel weight
- Courier tracking number entered before dispatch
- Fulfillment status updated to **shipped**
- Customer notified via automated WhatsApp/email

COMPREHENSIVE OPERATIONAL TROUBLESHOOTING GUIDE

SYMPTOM / ERROR	PROBABLE CAUSE	ADMINISTRATIVE CHECK	RESOLUTION PROCEDURE
Product not visible on storefront	Draft status or no category assigned	Check status toggle and category tag	Set toggle to "Active Storefront" and ensure at least one active category is selected.
Variant selector empty on product page	Variant values inactive or matrix not generated	Check Variants tab and Combination Matrix	Enable active toggles for all variant values and click "Generate Combination Matrix".
Default variant shows "Sold Out"	Default variant combination has zero stock	Check matrix row with the star icon	Allocate stock directly to the default row or move the star to an in-stock variant.
Photos not switching on swatch click	Images not linked to specific variant values	Open product Images tab and check dropdowns	Map each uploaded image to its corresponding variant value from the dropdown and save.
Incorrect shipping fee at checkout	Product weight missing or rules misconfigured	Check product weight field and global settings	Enter accurate weight in grams; review shipping threshold under Settings → Shipping.
Inbound WhatsApp messages missing	Webhook disconnected or Meta token expired	Check WhatsApp connection status in Settings	Verify Meta WhatsApp Cloud API credentials and check webhook health status.
Checkout blocked: "Out of Stock"	Variant-level stock is zero even if base stock > 0	Check Combination Matrix stock column	Allocate stock directly to the specific variant combination in the matrix table.
Database backup fails to complete	Storage bucket permission or quota issue	Check backup status widget in Settings	Verify Firebase Storage service account permissions on <code>db_backups/</code> .

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DOCUMENT HANDOVER & ACCEPTANCE

This verification checklist confirms that all administrative modules have been fully audited, tested, and handed over for live commercial operations.

OPERATIONAL VERIFICATION ITEM	VERIFICATION STATUS	OPERATIONAL NOTES
Admin authentication and session security verified	VERIFIED	Credentials scrubbed; no plaintext storage
Product catalog and category taxonomy verified	VERIFIED	All active SKUs mapped to categories
Variant engine and combination matrix verified	VERIFIED	Multi-group permutations tested
Variant templates tested across product types	VERIFIED	Filament, weight, and printer templates active
Weight-bracket courier calculation engine verified	VERIFIED	Real-time shipping tariffs calibrated
Order processing and inspection modals verified	VERIFIED	Full order lifecycle operational
Custom STL slicing quotes workflow verified	VERIFIED	Volume, mass, and infill parameters verified
Offline draft booking and recovery blasts verified	VERIFIED	Manual sales dockets and coupon triggers active
WhatsApp Business Cloud API inbox verified	VERIFIED	Inbound/outbound sync, reactions, and attachments active
WhatsApp rule-based auto-replies verified	VERIFIED	Greeting and keyword rules active (Zero-LLM)
Push Notification Hub and broadcast builder verified	VERIFIED	FCM integration active with deep-link routing
Customer review moderation workflow verified	VERIFIED	Approve/Reject actions functional
Database backup, SHA-256 hashing, and restore verified	VERIFIED	Automated weekly cron and snapshot restoration active



DELIVERED BY

AJR Digital Hub Team
Enterprise Digital Solutions Team

ACCEPTED BY

3D Galaxy Team
E-Commerce Operations & Store Administration

Signature & Date

Signature & Date